Parameter-Efficient Continued Pre-training and Instruction Tuning of a Medical Vision-Language Model

Course project, Topic 6 (Task 1.3) · Draft v1.6, 2026-10-25 · Author: LOO CHUN QIAN

> **Draft status.** Sections 3, 4, 5 and 6 are written from the repository as it stands. Experiments 0, A, B1, B2, C1, the A-server control and the full C2 replay curve (0 / 99 / 300 / 900, two seeds at 300 and 900) are complete, and the central replay experiment has been repeated on a 35B mixture-of-experts model at four replay budgets (0 / 99 / 300 / 900) with two seeds each, plus six group-D ablations and a free-text probe (Section 5.7), and evaluated on four external test sets approved by the instructor (Section 5.8). Experiment B3 (Section 5.9) has run: image-text CPT on CheXpert Plus with and without alignment-score filtering. Missed abnormalities are measured on 3,189 adult CheXpert Plus films (Section 5.10) and re-read by a model-based judge (Section 5.11). Three further 35B runs are complete: one SFT mix covering six datasets (Section 5.12), report-only fine-tuning on CheXpert Plus (Section 5.13) and the three-stage pipeline of knowledge pre-training, instruction tuning and reasoning training (Section 5.14); A and B have also been retrained and evaluated with SFT seed 43, confirming no reproducible gain from knowledge pre-training while leaving the effect of Chinese dialogue on adult-film omission unresolved. Section 2 (Related Work) is written; the 29 original entries in `docs/refs.bib` were checked against arXiv or the publisher page on 2026-09-23 and the nine added for Section 5.7 on 2026-09-27 (record in the file header). No `[TODO]` markers remain; open items are listed as limitations in Section 7. Repository: https://github.com/AugustLoo/MedLoRA

Abstract

Open-weight vision-language models (VLMs) in the 2–7B range answer general visual questions well but lag on medical images. This project studies a parameter-efficient adaptation pipeline, continued pre-training (CPT) followed by LoRA/QLoRA supervised fine-tuning (SFT), for Qwen2.5-VL-3B-Instruct, with three fixed evaluation tables: medical VQA (SLAKE), general-ability retention (TextVQA subset) and answer reliability (PubMedQA). QLoRA SFT on 4.9k SLAKE questions raises closed-question accuracy from 67.3 to 85.1 and open-question recall from 46.7 to 82.2 with no measurable loss on TextVQA. A text-only CPT stage on 10k PubMed abstracts does not transfer to image questions (SLAKE unchanged within ±0.6) but improves text-only medical reasoning (PubMedQA +2.2). An image-text CPT stage on 3.5k IU X-Ray image-report pairs learns the radiology report style but not the findings, and leaves SLAKE unchanged (closed 84.1) while lowering open lesion questions; in this regime, a 3B model with a frozen vision tower and a few thousand CPT examples, the CPT stage does not contribute to the primary metric. Evaluating the CPT-stage adapters without SFT localises a persistent "yes" bias to the short-answer SFT data rather than to CPT or to the base model. Acting on that diagnosis, mixing 900 three-class text examples into the SFT set (experiment C1) raises PubMedQA macro-F1 from 51.5 to 61.1 and triples the number of correctly answered "maybe" questions, at a cost of 2.2 points of general VQA accuracy and roughly one net SLAKE question; a single-variable control run on the same GPU isolates that trade-off to the replay data itself. Varying the replay budget over 0 / 99 / 300 / 900 examples, with two seeds at 300 and 900, shows the calibration gain saturates early: 99 examples buy 63 % of the 900-example gain and 300 buy 92 %, the remainder being within seed spread, at a smaller retention cost and no main-task cost. The per-class balance of the fix, by contrast, improves monotonically with budget: every replay budget over-corrects toward "no" and "maybe", and 900 examples bring the predicted distribution closest to the truth. A second, broader retention probe (MMBench, 500 fixed multiple-choice questions) shows no replay cost at all, which localises the measured

loss to short-answer output format rather than to visual ability. The alignment problem in this pipeline is therefore a data-mixture problem, addressable without a separate preference-optimisation stage. Repeating the zero- and 300-example conditions on Intern-S2-Preview, a 35B mixture-of-experts model roughly twelve times larger, with two seeds each, shows that the fix transfers but the cause does not: the 35B base model is as reluctant to answer "maybe" as the 3B base, short-answer SFT does not reliably make it worse, and 300 replay examples still raise macro-F1 by five points on average (61.4 → 66.3) without the over-correction seen at 3B and at no measurable cost to the main task. Extending the 35B budget to 99 and 900 examples reproduces the shape of the 3B curve — the smallest budget over-predicts "maybe" at both scales — but flattens it: every budget from 99 to 900 lies within seed spread of the others (macro-F1 64.7–66.3), and none costs retention. Letting the 35B base model reason before answering does not substitute for replay: on the questions it finishes within 4,096 tokens it answers 5 of 37 "maybe" questions correctly against 9-12 for the 300-example replay models, and it fails to finish on a third of the "maybe" questions. Ablating the 35B training settings shows that only under-training reliably hurts and that LoRA on the shared expert can be dropped; a free-text probe shows no degradation of long-form description. On four external test sets the fine-tuned 35B models gain 2.8–4.8 points on VQA-RAD, lose 2.3–3.9 points on PathVQA while shifting toward "no", and lose 1.0–1.6 points on MedQA, consistently across four runs; replay makes no measurable difference there, and on chest X-rays every model often describes a film it has just called pneumonia as free of lung abnormality. Image-text CPT on 5,000 CheXpert Plus pairs, chosen either by a Topic 1 alignment score or at random, changes neither the main task nor the comparison between the two (B3). On 3,189 adult chest X-rays every fine-tuned 35B model answers "abnormal" more readily when asked (missed abnormal films 12.9 % → 2–5 %, specificity 82 % → 38–59 %), while three of four runs describe more abnormal films as normal (29 % → 53–69 %), so the two question formats increasingly contradict each other; re-reading every description with CheXbert keeps the ordering of models and confirms the seed dependence. Training on the training splits of all six evaluation datasets leaves the main tables unchanged, lifts PathVQA by 12.5 points over the base and, without any description training, lowers description omission on both paediatric and adult chest films (adult: 11.2 % by CheXbert, one seed), while report-only training on 10,000 CheXpert Plus pairs gives the most balanced adult-film behaviour (omission 10.2 %, false alarms 25.1 %) and nothing else. Finally, the full three-stage pipeline at 35B (19.4 M tokens of textbook, guideline and encyclopedia pre-training, then 27,953 instruction examples, then 13,000 explained exam questions) shows that the gains come from instruction tuning: across SFT seeds 42 and 43, knowledge pre-training gives no reproducible measurable improvement, with a small PathVQA closed decline at both seeds; reasoning training has only a single-seed PneumoniaMNIST gain. Changing the SFT seed alone shifts adult-film omission by about four points, so the effect of Chinese dialogue remains unresolved under the keyword judge pending CheXbert re-reading. Chinese dialogue data does not lift Chinese short open answers, and a no-image control confirms the models answer from the image. Data-generation scripts, evaluation code and configurations are released for one-command reproduction; the trained adapters are archived by the author rather than published.

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1. Introduction

Large VLMs such as Qwen2.5-VL and LLaVA are trained on web-scale natural images. Radiology and pathology images are rare in that data, so out of the box these models mislabel organs, confuse modalities and hedge poorly on medical questions. Full fine-tuning of even a 3B model is out of reach for a single consumer GPU, which motivates parameter-efficient methods: low-rank adapters (LoRA) and their 4-bit variant (QLoRA) train under 2% of the parameters and fit on a 16 GB card.

Task 1.3 of the course asks for a complete pipeline on a 2–7B open VLM: (1) continued pre-training on medical corpora, (2) LoRA/QLoRA instruction tuning, (3) an alignment step, with four deliverables: data-generation scripts, the CPT/SFT pipeline with adapters, evaluation sets with model cards, and a report with reproducible configurations.

This report makes the following contributions:

1. A reproducible three-table evaluation protocol (medical VQA, general retention, reliability) shared by every experiment, with prompts fixed once and used identically for training data generation and evaluation.
2. A controlled comparison of SFT alone versus CPT followed by SFT, on the same base model, adapter rank, learning schedule and evaluation seed.
3. Evidence that text-only CPT does not transfer to image questions, motivating image-text CPT on chest X-ray reports (IU X-Ray, then CheXpert Plus).
4. A question-type error analysis that separates "vocabulary alignment" gains from genuine gains in medical perception.

The rest of the report is organised as follows. Section 2 reviews related work. Section 3 describes the pipeline. Section 4 gives the datasets and experimental setup. Section 5 reports results, Section 6 discusses them, Section 7 lists limitations and planned work, and Section 8 gives reproduction instructions.

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2. Related Work

Open vision-language models at the 2–7B scale. Contemporary open VLMs share an architecture: a ViT vision tower, a projector that maps visual features into the language model's embedding space, and a decoder-only LLM. LLaVA [30] established visual instruction tuning as the standard recipe, and the Qwen2-VL / Qwen2.5-VL line [2, 45] added dynamic-resolution visual tokens, which matters here because it lets a 3B model take a 512×512 radiograph without a fixed-grid resize. Earlier few-shot multimodal work such as Flamingo [1] used frozen backbones with cross-attention adapters; the projector-based design has since become dominant because it fine-tunes more cheaply. This project takes Qwen2.5-VL-3B-Instruct as given and asks what can be added to it under a 16 GB memory budget.

Medical vision-language models. LLaVA-Med [24] is the closest reference point: it adapts LLaVA to biomedicine in two stages — caption-style alignment on PMC figure-caption pairs, then instruction tuning on generated biomedical dialogues. Med-Flamingo [33] pursues few-shot medical reasoning from interleaved textbook data, and Med-PaLM M [43] scales a generalist biomedical model far beyond the compute available here. **Our experiment B2 is a deliberately small-scale reproduction of LLaVA-Med's first stage**: caption loss over image-report pairs with a frozen vision tower. The result — the adapter learns the report template and not the findings — is reported as a negative result about that stage at this scale, not as a criticism of the original work, which uses two orders of magnitude more data and trains the projector.

Parameter-efficient adaptation. LoRA [15] injects trainable low-rank matrices into frozen weights; QLoRA [10] adds a 4-bit NF4 quantised base and paged optimisers, bringing 7B-scale fine-tuning within a single consumer GPU. Both are reported to approach full fine-tuning on instruction-following tasks at a fraction of the memory. All training here runs through LLaMA-Factory [49], with 0.79 % of parameters trainable and the vision tower frozen throughout — a choice that makes every measured change attributable to the language model, and that also bounds what the CPT stages could possibly achieve.

Domain-adaptive continued pre-training. Gururangan et al. [13] showed that continued pre-training on in-domain text reliably helps in-domain downstream tasks, and domain-specific encoders such as PubMedBERT

[11] make the same case for biomedicine. That literature is almost entirely unimodal. **The question this project actually tests is whether the finding survives a modality gap**: does continued pre-training on biomedical *text* improve a *visual* question-answering task? Experiments B1 and B2 answer no at this scale, while confirming that text CPT does help the text-only reliability table — the gain stays inside its own modality.

Forgetting and replay. Rehearsal — mixing samples from the original distribution into new training — is the oldest remedy for catastrophic forgetting [39], alongside regularisation approaches such as EWC [22]. LoRA is often argued to forget less than full fine-tuning simply because it changes less. This project uses replay for an unusual purpose: not to preserve a previous *task*, but to preserve a previous *answer format*. The three-class PubMedQA examples mixed into the SFT set (experiments C1, C2) exist to stop the model losing the ability to say "maybe", and the general-ability probe then measures what that costs.

Alignment and calibration. The standard third stage after instruction tuning is preference optimisation, whether RLHF [34] or the simpler DPO [38]. Separately, a long line of work documents that neural networks are poorly calibrated [12] and that language models have some latent, recoverable sense of their own uncertainty [21]. **This project's main finding sits between those two literatures**: the reliability failure it measures is not inherited from pre-training but *manufactured by the instruction-tuning data*, and correcting the stage-2 data mixture recovers most of it without a preference-optimisation stage at all. Section 5.5 quantifies both the recovery and its cost.

Benchmarks and data. Medical VQA is evaluated on SLAKE [29], chosen over VQA-RAD [23] and PathVQA [14] for its size, its English subset and its question-type annotations, which make the error analysis in Section 5.2 possible. Reliability is measured on PubMedQA [18], whose three-way {yes, no, maybe} label is what makes hedging measurable at all. General-ability retention uses TextVQA [41]; its limitations as a probe are discussed in Section 6, and MMBench [28] is the planned replacement. Image-report pairs come from IU X-Ray / OpenI [9], with CheXpert Plus [4] — the report-augmented extension of CheXpert [17] — registered for the scale-up; MIMIC-CXR [20] was in the original task description but was replaced for access reasons (Section 4.1). Model and data documentation follows the model-card convention [32].

Position of this work. Nothing here is a new method. The contribution is a controlled measurement on one small open model under a fixed compute budget: three evaluation tables held constant across eight adapters, a single-variable control run to separate hardware from data effects, per-question predictions stored so every number is recomputable, and one negative result (CPT does not cross the modality gap at this scale) reported as carefully as the positive one (the calibration failure is a data-mixture artefact and is fixable as such).

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3. Method

3.1 Pipeline overview

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base model ──► (stage 1) CPT adapter ──► (stage 2) SFT adapter ──► (stage 3) alignment  [planned]
Qwen2.5-VL-3B     medical corpus           SLAKE train                 preference data
                  text or image-text       instruction format          (yes/no/maybe calibration)
```

Each stage produces a LoRA adapter on the same frozen 4-bit base. Stage 2 is initialised from the stage-1 adapter (`adapter_name_or_path`) and continues training the same low-rank matrices, so the final artefact is a single 120 MB adapter. Experiment A skips stage 1; experiments B1/B2 differ only in the stage-1 corpus.

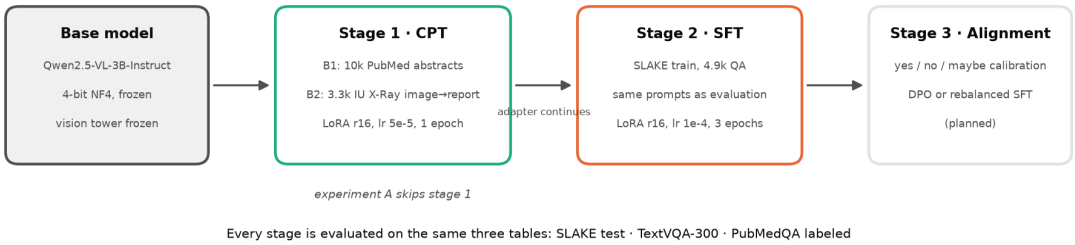


Figure 1. The three-stage pipeline. Every adapter is evaluated on the same three tables.

3.2 Base model

Qwen2.5-VL-3B-Instruct (Apache-2.0): a 3.75B-parameter model with a native-resolution ViT, an MLP projector and a Qwen2.5 decoder. It was chosen because (i) it sits in the 2–7B range required by the task, (ii) it fits on a 16 GB T4 in 4-bit with room for activations, and (iii) it already follows short-answer instructions, which makes zero-shot baselines meaningful.

3.3 Parameter-efficient training (QLoRA)

- Base weights quantised to 4-bit NF4 with double quantisation (bitsandbytes); LoRA matrices and optimiser states in fp16.
- LoRA on every linear layer of the language model (`lora_target: all`), rank 16, α = 32, dropout 0.05. The vision tower and projector are frozen in all experiments, so any change in behaviour is attributable to the LLM side.
- Gradient checkpointing, effective batch 16 (2 × 8 accumulation), cosine schedule, seed 42. Training is done with LLaMA-Factory; one YAML per experiment under `configs/`.

3.4 Stage 1a: text-only CPT (B1)

Causal-LM loss on 10,000 PubMedQA `pqa_artificial` documents (question + abstract context + long answer, concatenated), packed into 1024-token sequences. Learning rate 5e-5 (half of SFT), one epoch. The evaluation split `pqa_labeled` is never used for training.

3.5 Stage 1b: image-text CPT (B2)

Caption-style alignment in the sense of LLaVA-Med stage 1: the input is a frontal chest X-ray plus a fixed instruction ("Write the radiology report for this chest X-ray."), the target is the report's Findings and Impression sections, and the loss is computed on the report tokens only. Implemented as an SFT-format dataset so that images are consumed; hyper-parameters as in 3.4. Corpus: IU X-Ray (see 4.1); CheXpert Plus is the planned scale-up.

3.6 Stage 2: instruction tuning (SFT)

SLAKE English training questions converted to a two-turn chat format with the same prompt templates used at evaluation time (`medvlm/prompts.py`):

- closed questions: `{question}\nAnswer with yes or no only.`
- open questions: `{question}\nAnswer with a single word or short phrase.`

Learning rate 1e-4, three epochs, images resized to at most 512×512 pixels (262,144), validation loss on the SLAKE validation split every 400 steps.

3.7 Stage 3: alignment

The PubMedQA results show a stable over-prediction of "yes" and under-prediction of "maybe". The original plan was DPO on preference pairs built from PubMedQA `pqa_artificial` (preferred: the gold label; rejected: the model's over-confident answer). Two findings changed that plan before it was implemented.

First, evaluating the CPT-stage adapters without SFT (Section 5.3) showed the bias is *introduced* by the short-answer SLAKE SFT data: the CPT-only adapters are the best-calibrated models in the whole project, better than the base model. The failure is created in stage 2, not inherited from pre-training, so a third stage would be correcting damage that a stage-2 data change could avoid. Second, the direct test of that hypothesis (experiment C1, Section 5.5) recovers most of the calibration by mixing 900 three-class text examples into the SFT set: macro-F1 51.45 → 61.09 with a measured cost of 2.2 points of general VQA accuracy.

Stage 3 is therefore reframed as *SFT data-mixture design* rather than preference optimisation, and the remaining question is quantitative: how much replay is enough, and how the calibration gain trades against retention. Experiment C2 varies the replay budget to trace that curve; Section 5.6 reports the 0 / 300 / 900 points. DPO is retained as an optional experiment E, to be run only if the data-mixture route saturates below acceptable calibration — which, on the evidence so far, it does not.

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4. Data and Experimental Setup

4.1 Datasets

Dataset	Role	Size used	Licence / access
SLAKE (English)	SFT training; validation loss; **test set for medical VQA**	train 4,919 / val 1,053 / test 1,061 (416 closed, 645 open)	CC BY 4.0
PubMedQA `pqa_artificial`	text CPT corpus	10,000 documents	MIT
PubMedQA `pqa_labeled`	**reliability evaluation** (yes/no/maybe); from experiment C1 on, a fixed stratified half is also a three-class SFT source	1,000, split 500 train / 500 test (seed 42, stratified)	MIT; the test half is never trained on
TextVQA validation	**general-ability retention** (short-answer)	fixed 300-question sample, seed 42	CC BY 4.0
MMBench (en, dev)	**general-ability retention** (multiple-choice, 20 ability dimensions)	fixed 500-question sample, seed 42	see dataset card
IU X-Ray (OpenI, Kaggle mirror)	image-text CPT prototype	≈3.3k frontal images with non-empty reports, 95/5 split by report id	public
CheXpert Plus	image-text CPT, B3 (and the teammate's test split, planned for missed-abnormality evaluation)	10,000-study pool; 5,000 per arm	registered; on the college server only

MIMIC-CXR was in the original task description; CheXpert Plus was chosen instead because it provides the same image-report structure with a lighter access process (registration versus PhysioNet credentialing), and because it is the dataset shared with the Topic 1 teammate whose image-text alignment model will provide the data-filtering scores for B3.

Compliance rules fixed in the repository: SLAKE test and PubMedQA labeled are evaluation-only; patient-level splits are stored in git; controlled data and patient-level derived files never enter git or public Kaggle datasets.

4.2 Evaluation protocol

All three tables are produced by `train/eval_all.sh <tag> <adapter>` with greedy decoding (`do_sample=False`), at most 32 new tokens, images capped at 512×512, on the full-precision fp16 base plus adapter. Predictions are stored per question for error analysis.

Table	Data	Metrics
Medical VQA	SLAKE test	closed: see the scoring note below; open: exact match, token recall (LLaVA-Med convention) and token F1, all after lower-casing, stripping punctuation and articles; both split by modality (X-Ray / CT / MRI)
General retention	TextVQA 300	VQA accuracy, min(#matching annotators / 3, 1)
Reliability	PubMedQA labeled, **held-out half only from C1 on**	accuracy, macro-F1 over {yes, no, maybe}, predicted-label distribution versus gold (full 1,000: 552 / 338 / 110; held-out 500: 276 / 169 / 55)
General retention (2nd probe)	MMBench en-dev, 500 fixed	accuracy on the option letter (first A–D in the output); no circular evaluation; per-category breakdown stored

Scoring note on closed questions (corrected 2026-09-19). SLAKE's CLOSED category is not purely yes/no: 61 of the 416 closed test questions have a closed-vocabulary gold answer such as *Lung*, *Liver*, *T2* or *Coronal Plane*. Our first implementation folded both prediction and gold through a yes/no/other mapping, so on those 61 questions any answer that was not literally "yes" or "no" collapsed to "other" and matched the gold, scoring as correct for free. This inflated every instruction-tuned adapter, which had learned to answer those questions with a content word, by 4 to 5 points; the zero-shot base was unaffected because it forced yes/no answers there and was scored correctly. Closed questions are now scored as yes/no agreement when the gold is yes/no, and as exact match otherwise (`medvlm.metrics.closed_score`). All numbers in Section 5 use the corrected scorer; `scripts/rescore_slake.py` recomputes them from the stored per-question predictions without re-running the models. The correction lowers experiment A from 89.18 to 85.10, B1 from 88.70 to 83.89 and B2 from 88.70 to 84.13, and leaves every conclusion unchanged.

PubMedQA split protocol (introduced for C1). Experiment C1 trains on PubMedQA text, so the reliability table had to be split before it could be used as both a training source and an evaluation set. `data/split_pubmedqa.py` draws a label-stratified 500 / 500 split with seed 42 and writes `data/pubmedqa_split.json`, which is committed to git and refuses to be regenerated once it exists. The training half contributes 276 yes / 169 no / 55 maybe to experiment C1; the held-out half has the same distribution and is never trained on. Because the earlier models were scored on all 1,000 questions, `scripts/eval_pubmedqa_split.py` recomputes every model's PubMedQA metrics on the held-out half from the stored per-question predictions, so all runs are compared on one ruler without re-running any model. Sections 5.1 and 5.3 quote the full-1,000 numbers for continuity with the earlier draft; Section 5.5 and every comparison involving C1 quote the held-out 500.

4.3 Hardware and cost

Experiments 0, A, B1 and B2 ran on Kaggle (one NVIDIA T4, 16 GB, fp16, 4-bit NF4 base). From experiment C1 on, training moved to the college GPU server (one RTX 5090, 32 GB, bf16, **no quantisation**); a local RTX 3050 (4 GB) was used only for code and smoke tests. The two platforms therefore differ in both numerical precision and quantisation, which is why the baseline was re-evaluated on the server and why experiment A was re-run there (Section 5.5): the baseline agrees to within one point on all five metrics and A-server reproduces A to within half a point, so figures from the two platforms can be read on the same axis.

Run	Samples	Wall time	Throughput
Baseline evaluation (3 tables)	–	≈1.5 h	SLAKE 1,061 questions in 38 min
A: SLAKE SFT, 3 epochs	4,919 × 3	4 h 28 min	0.92 samples/s
B1: text CPT, 1 epoch	10,000 packed	2 h 39 min	0.42 samples/s

Run	Samples	Wall time	Throughput
B1: SFT after CPT	4,919 × 3	4 h 07 min	1.0 samples/s
B2: image-text CPT, 1 epoch	3,483 image-report pairs	1 h 19 min	0.73 samples/s
B2: SFT after CPT	4,919 × 3	4 h 38 min	0.89 samples/s
(below: RTX 5090, bf16, unquantised)			
Baseline re-evaluation (3 tables)	–	≈1 h	–
C1: SLAKE + PubMedQA replay SFT, 3 epochs	5,819 × 3	1 h 25 min	3.42 samples/s
A-server: SLAKE SFT, 3 epochs	4,919 × 3	6 h 19 min	0.65 samples/s
C2-300: SLAKE + 300 replay SFT, 3 epochs	5,219 × 3	10 h 58 min	0.40 samples/s
C2-300-s43: same, seed 43	5,219 × 3	**45 min**	5.7 samples/s
(below: Intern-S2-Preview 35B, 4 × RTX 5090, bf16, layer-wise model parallel)			
S2-300: SLAKE + 300 replay LoRA SFT, 3 epochs	5,219 × 3	1 h 58 min	2.2 samples/s
S2-0: SLAKE LoRA SFT, 3 epochs	4,919 × 3	1 h 18 min	3.2 samples/s
S2-99: SLAKE + 99 replay, 3 epochs, seeds 42 / 43	5,018 × 3	1 h 52 min / 1 h 58 min	2.1–2.2 samples/s
S2-900: SLAKE + 900 replay, 3 epochs, seeds 42 / 43	5,819 × 3	2 h 12 min / 2 h 19 min	2.1–2.2 samples/s
C2-100: SLAKE + 99 replay SFT, 3 epochs	5,018 × 3	not recorded	–
C1-s43: SLAKE + 900 replay SFT, seed 43	5,819 × 3	not recorded	–

The A-server and C2-300 runs are eight times slower per optimiser step than C1 on the same GPU (38–40 s versus 4.7 s) with the GPU idle 95 % of the time; the second-seed run C2-300-s43 then took 45 minutes (2.8 s per step). Across the four server runs, the slow ones coincide with a job belonging to another account saturating both GPUs and the fast ones with that job idle, so the slowdown is attributed to GPU time-sharing (a correlation across four runs, not a proof) and is recorded as an open operational issue in `docs/SERVER.md` rather than as a property of the method. It does not affect the trained weights: A-server reproduces A's validation loss to four decimal places.

4.4 Experiment matrix

ID	Stage 1	Stage 2	Question answered	Status
0	–	–	zero-shot starting point	done
A	–	SLAKE SFT	gain from instruction tuning alone	done
B1	text CPT (PubMedQA)	SLAKE SFT	does text CPT transfer to image VQA	done
B2	image-text CPT (IU X-Ray)	SLAKE SFT	does image-report CPT transfer	done
B1/B2 CPT-only	stage-1 adapters, no SFT	–	what CPT itself changes	done
B3	image-text CPT on CheXpert Plus: top-5k by alignment score vs random-5k from 10k unseen pairs	SLAKE SFT	value of data-quality filtering	done (§5.9)
MISS	–	missed-abnormality metric on the CheXpert Plus test split (3,189 adult films, CheXbert labels), yes/no and free description, five 35B models	does fine-tuning hide abnormalities	done (§5.10)

ID	Stage 1	Stage 2	Question answered	Status
A-server	–	SLAKE SFT, bf16 on the 5090	single-variable control for C1; does A survive the platform change	done
C1	–	SLAKE SFT + 900 PubMedQA three-class examples	can the SFT data mix fix the "yes" bias, and what does it cost	done
C2	–	as C1 with 99 / 300 / 900 replay examples (0 = A-server)	how much replay is enough; shape of the trade-off	0, 99, 300, 900 done; 1,800 deprioritised (§5.6)
C2-s43	–	C2-300 and C1 (900) each repeated with sampling and training seed 43	run-to-run variance at both ends of the curve	done (§5.6)
S2-0 / S2-300	–	SLAKE SFT with 0 / 300 replay examples on Intern-S2-Preview (35B MoE), two seeds each	do the 3B findings hold at 12× the parameters	done (§5.7)
S2-99 / S2-900	–	S2 with 99 / 900 replay examples (the 3B C2 files), two seeds each	does the 3B budget curve hold at 35B	done (§5.7)
D (35B)	–	S2-300 with one change each: 1 epoch; attention-only LoRA; rank 8 / 32; lr 5e-5 / 2e-4; seed 43 for control, attention-only and 1 epoch	which training factor matters	done (§5.7)
OE	–	free-text probe: COCO descriptions and SLAKE free-text reports, all thirteen 35B models	does long-form generation drift	done (§5.7)
EXT	–	VQA-RAD, PathVQA, MedQA, PneumoniaMNIST test sets; 35B base and replay 0 / 300, two seeds each	transfer, out-of-domain yes/no bias, knowledge retention, missed abnormalities	done (§5.8)
D (3B)	ablations: rank 8/16/32, lr, epochs		the same factors at 3B for comparison	planned, if time allows
E (optional)	C + DPO		reliability	deprioritised: C1 shows the data mix alone recovers calibration

5. Results

5.1 Main table

Metric	Base (0)	A: SFT	B1: text CPT → SFT	B2: image CPT → SFT
SLAKE closed acc	67.31	**85.10**	83.89	84.13
SLAKE open EM	40.62	75.35	**75.81**	74.57
SLAKE open recall	46.73	82.17	**82.72**	81.61
SLAKE open F1	47.53	81.56	**82.07**	80.88
SLAKE closed, X-Ray / CT / MRI	80.70 / 64.49 / 56.82	85.09 / 84.58 / 86.36	82.46 / 83.64 / 86.36	85.09 / 82.71 / 86.36
TextVQA acc (retention)	83.89	83.89	83.78	84.00
PubMedQA acc	65.80	70.60	**72.80**	70.80
PubMedQA macro-F1	51.03	52.38	**54.19**	52.96
PubMedQA predicted yes / no / maybe (gold 552 / 338 / 110)	641 / 201 / 158	685 / 256 / 59	670 / 286 / 44	672 / 263 / 65

Training curves: A, train loss 0.736 → 0.082, validation loss 0.225 → 0.145 still decreasing at epoch 3; B1 CPT loss 1.96 → 1.75 over one epoch; B1 SFT validation loss 0.160 → 0.142; B2 CPT loss 2.17 → 1.06; B2 SFT validation loss 0.168 → 0.151. No NaN in any run.

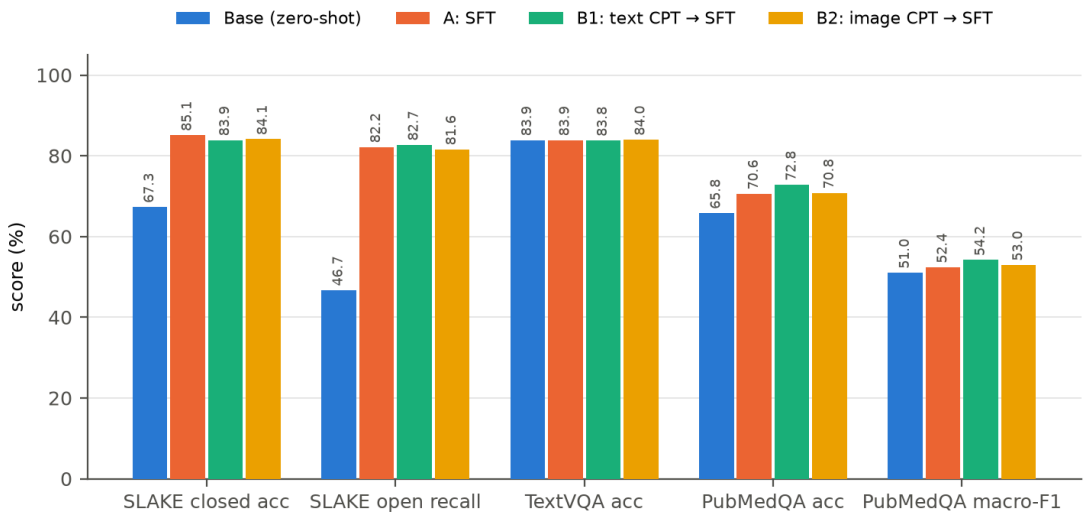


Figure 2. Main results across the three evaluation tables.

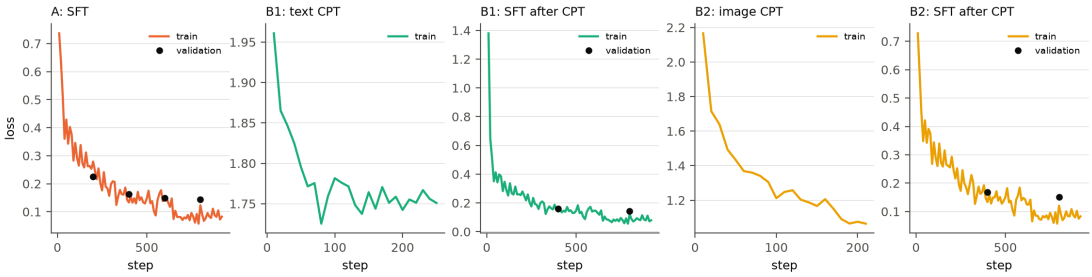


Figure 4. Training loss (line) and validation loss (dots) for each run.

5.2 Results by question type (SLAKE test, accuracy; open questions use exact match)

Type	n	Base	A	B1	B2	B1 – A	B2 – A
Position, open	163	24.5	57.7	60.1	58.3	+2.5	+0.6
Organ, closed	154	75.3	90.9	90.3	89.6	−0.6	−1.3
Abnormality, closed	109	68.8	78.0	75.2	78.9	−2.8	+0.9
Knowledge-graph, open	109	22.0	72.5	69.7	70.6	−2.8	−1.8
Organ, open	99	23.2	84.8	83.8	84.8	−1.0	0.0
Modality, open	75	92.0	94.7	94.7	94.7	0.0	0.0
Quantity, open	52	71.2	73.1	76.9	75.0	+3.8	+1.9
Abnormality, open	41	12.2	41.5	39.0	**31.7**	−2.4	**−9.8**
Modality / Plane / Colour / Size, closed	91	26.9–78.6	84.6–100.0	84.6–100.0	84.6–100.0	0.0	−1.4

Per-question changes: base → A fixed 328 questions and broke 30; A → B1 fixed 15 and broke 17; A → B2 fixed 13 and broke 22. On the X-ray subset alone (the CPT modality), B2 versus A is 74.5 versus 76.5 on open questions and 85.1 versus 85.1 on closed.

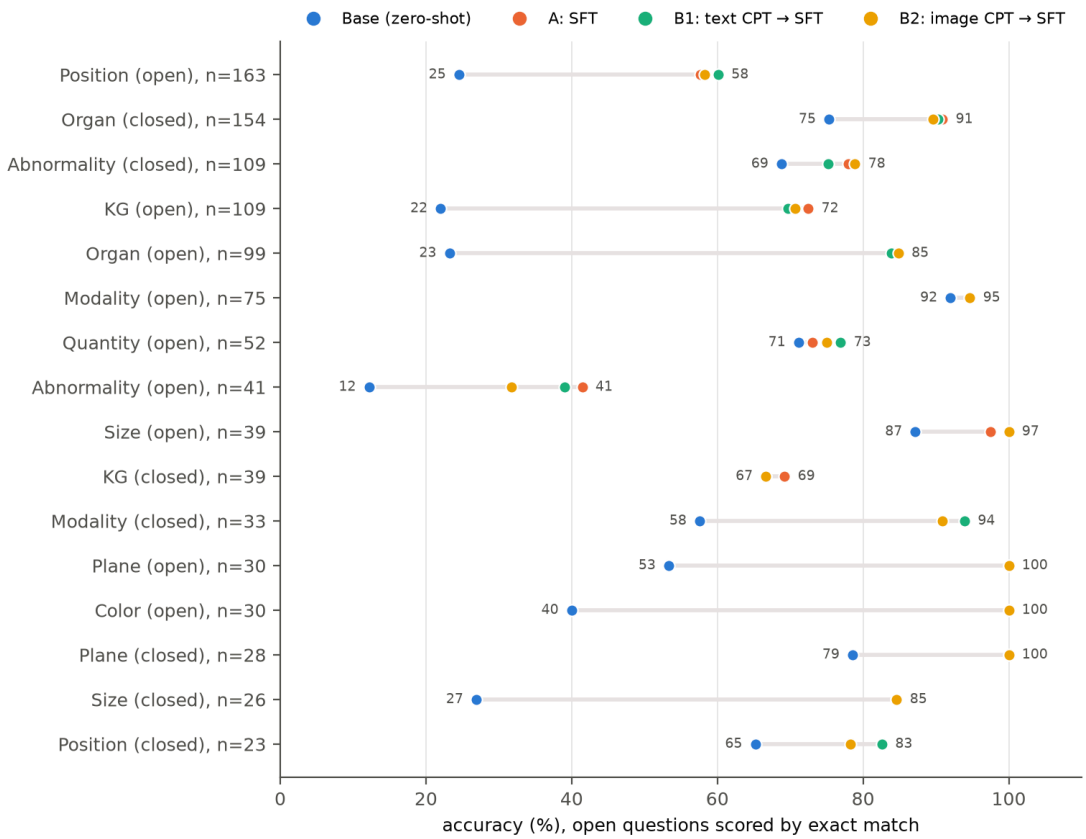


Figure 3. SLAKE test accuracy by question type. The grey bar spans base → A; the largest gains are on vocabulary-bound types (Organ, KG, Colour, Plane), the smallest on Abnormality and Position.

5.3 CPT-stage adapters evaluated without SFT

To separate what the CPT stage itself does from what the SFT stage overwrites, the two stage-1 adapters were evaluated directly, without stage 2.

Metric	Base	B1 CPT only	B2 CPT only	A: SFT
SLAKE closed acc	67.31	69.71	68.27	85.10
SLAKE open EM / recall	40.62 / 46.73	34.11 / 44.87	36.43 / 44.65	75.35 / 82.17
TextVQA acc	83.89	81.67	83.56	83.89
PubMedQA acc / macro-F1	65.80 / 51.03	66.70 / 53.25	68.40 / 53.20	70.60 / 52.38
PubMedQA predicted yes / no / maybe (gold 552 / 338 / 110)	641 / 201 / 158	578 / 251 / 171	**598 / 288 / 114**	685 / 256 / 59

Three things follow. (i) CPT is not inert: closed accuracy rises by 2.6–3.1 points and PubMedQA by 0.9–2.6, but the gains are small and are entirely absorbed by the SFT stage, after which all adapters converge to the same scores. (ii) The drop in open-question exact match is not verbosity: CPT-only answers average 1.3–1.4 words, the same as the base. It is vocabulary drift concentrated in a few types (B1 CPT: Modality-open 92.0 → 53.3; B2 CPT: Plane-open 53.3 → 13.3, e.g. "Chest" answered as "Thorax"), which the SFT stage then realigns to the SLAKE vocabulary. (iii) The CPT-only models are the best-calibrated on PubMedQA: the B2 CPT adapter's label distribution (598 / 288 / 114) is the closest of all five models to the gold distribution. The "yes" bias and the collapse of "maybe" are therefore introduced by the short-answer SLAKE SFT, not by CPT and not by the base model. This relocates the alignment problem from stage 3 to the stage-2 data mix.

5.4 Experiment B2: what the image-text CPT stage learned

Before the SFT stage, the B2 CPT adapter was asked to write reports for three held-out IU X-Ray images. The generations are fluent, correctly structured radiology reports ("Findings: The heart is normal in size. The lungs are clear. No pneumothorax or pleural effusion. Impression: No acute cardiopulmonary abnormality."), but two of the three reference reports describe abnormalities (bibasilar opacities; a right pleural opacity with effusion) and both were reported as normal. The CPT loss fell from 2.17 to 1.06 in one epoch, so the objective was learned; what was learned is the dominant report template. In the IU X-Ray training split 36 % of reports are normal, and abnormal reports are still written mostly as negations ("no effusion"), so a caption loss with a frozen vision tower rewards the template rather than the image.

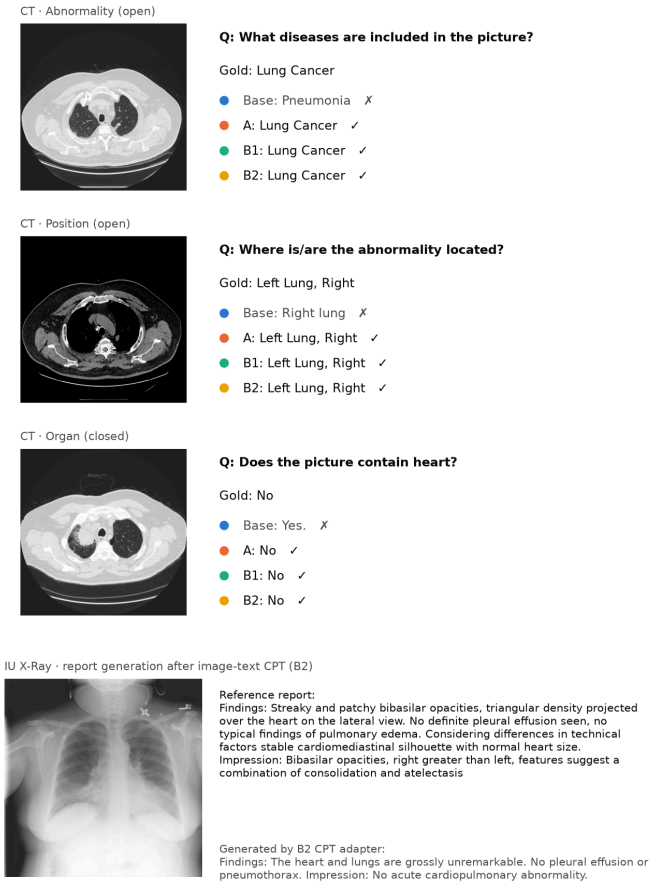


Figure 6. Top: three SLAKE test questions the base model gets wrong and every fine-tuned model gets right. Bottom: an abnormal IU X-Ray held-out image; the B2 CPT adapter writes a fluent normal report.

5.5 Experiment C1: three-class replay in the SFT mix, with a single-variable control

Section 5.3 located the "yes" bias in the short-answer SFT stage rather than in CPT or the base model. C1 tests that diagnosis directly: keep experiment A's recipe unchanged and add 900 three-class text examples (300 per label, sampled with replacement from the 500-question training half; "maybe" has only 55 unique items and is repeated about five times) to the 4,919 SLAKE questions. Every other hyper-parameter matches A — rank 16, lr 1e-4, three epochs, effective batch 16, seed 42, frozen vision tower.

Because A was trained on Kaggle in 4-bit fp16 and C1 on the server in unquantised bf16, the pair differed in three things at once. **A-server** removes two of them: experiment A's configuration re-run on the server under C1's exact conditions, leaving the 900 examples as the only difference.

A-server reproduces A. Across the platform change the two runs agree to within half a point everywhere, including the validation loss:

Metric	A (Kaggle, 4-bit fp16)	A-server (5090, bf16)	Δ
SLAKE closed / open EM / open recall	85.10 / 75.35 / 82.17	85.34 / 75.50 / 82.07	+0.24 / +0.15 / −0.10
TextVQA	83.89	83.56	−0.33
PubMedQA macro-F1 (held-out 500)	51.69	51.45	−0.24
Final validation loss	0.145	0.1446	−0.0004

This validates reading Kaggle and server figures on one axis, and it independently reproduces Section 5.3's finding: A-server makes 63 "no"→"yes" errors against the server baseline's 42, so the SFT stage pushes the model towards "yes" regardless of hardware or quantisation.

What the replay buys and what it costs. All PubMedQA figures below are on the held-out 500 questions (gold 276 / 169 / 55); SLAKE and TextVQA are the full test sets.

Metric	Server base	A-server	C1	C1 – A-server
PubMedQA macro-F1	48.66	51.45	**61.09**	**+9.63**
PubMedQA accuracy	64.00	68.80	71.60	+2.80
"maybe" answered correctly (of 55)	7	6	**19**	**+13**
"no" answered as "yes"	42	63	**26**	**−37**
Predicted yes / no / maybe	320 / 97 / 83	356 / 112 / 32	**269 / 151 / 80**	gold 276 / 169 / 55
SLAKE closed	66.35	85.34	84.13	−1.21
SLAKE open EM / recall / F1	40.93 / 47.19 / 47.92	75.50 / 82.07 / 81.19	76.12 / 82.19 / 81.32	+0.62 / +0.12 / +0.13
SLAKE closed, X-Ray / CT / MRI	78.95 / 64.95 / 53.41	84.21 / 83.18 / 92.05	83.33 / 83.18 / 87.50	−0.88 / 0.00 / −4.55
TextVQA	84.22	83.56	**81.33**	**−2.23**

Three readings follow.

The calibration gain is real, not a shifted prior. C1's macro-F1 rises 9.6 points, the largest single-intervention move in the project, and its predicted distribution (269 / 151 / 80) is the closest of any model to gold. The gain is not bought by guessing "maybe" more often: the base model also answers "maybe" 83 times but is right 7 times (8.4 % precision), while C1 answers it 80 times and is right 19 (23.8 %) — comparable volume, roughly three times the precision, and recall up from 12.7 % to 34.5 %. Precision and recall improve for all three labels simultaneously, and the "no"→"yes" error count falls below the untuned base model's.

SLAKE is essentially unchanged. Closed accuracy loses 1.21 points (about five of 416 questions, concentrated in MRI) and open exact match gains 0.62 (about four of 645), for a net change of roughly one question across the 1,061-question test set. C1's validation loss during training (0.142) is also marginally better than A-server's (0.1446). The earlier claim of "no cost on SLAKE" is therefore slightly too strong and is restated here as a small closed-question cost offset by a small open-question gain.

The cost lands on general ability, and the control attributes it. TextVQA decomposes cleanly: the server baseline scores 84.22, SLAKE SFT alone (A-server) takes it to 83.56 (−0.66, noise level), and adding the replay takes it to 81.33 (a further −2.23). The 2.9-point drop first observed for C1 against the baseline is therefore **not** caused by instruction tuning; it is caused by mixing text-only examples into a vision-language SFT set. Per question, 14 of 300 get worse and 6 get better, and the average answer length is unchanged (1.39 → 1.43 words), so this is not verbosity: about half the regressions are surface variants ("one penny" → "1 penny") and about half are genuine misreads ("Vietnam" → "us army").

The trade is explicit: 2.2 points of general VQA accuracy for 9.6 points of three-class macro-F1, a tripling of correct "maybe" answers, and 37 fewer false "yes" answers on clinical questions. Whether that exchange rate is favourable depends on the deployment; what matters methodologically is that it is now measured rather than assumed. Section 5.6 turns the single point into a curve.

5.6 Experiment C2: how much replay is enough

C1's 900 examples were a guess. C2 varies the budget with everything else fixed; the A-server run is the zero-replay point, so three points were available once a 300-example run finished. The 300-example sample is 100 per class from the same training half, sampled with replacement (the 55 unique "maybe" items repeat about 1.8×, versus 5.5× at 900). PubMedQA figures are on the held-out 500; SLAKE and TextVQA are full test sets.

Replay	macro-F1	acc	"maybe" correct	predicted yes / no / maybe	SLAKE closed	open EM	TextVQA
0 (A-server)	51.45	68.80	6 / 55	356 / 112 / 32	85.34	75.50	83.56
300	**60.48**	69.40	**21 / 55**	228 / 178 / 94	**85.82**	75.97	**82.33**
900 (C1)	61.09	71.60	19 / 55	269 / 151 / 80	84.13	76.12	81.33

Replay	yes P / R	no P / R	maybe P / R	"no"→"yes"	"yes"→"no"	dangerous flips
0	71.1 / 91.7	75.9 / 50.3	18.8 / 10.9	63	18	81
300	84.6 / 69.9	74.7 / 78.7	22.3 / 38.2	**14**	32	46
900	81.8 / 79.7	78.8 / 70.4	23.8 / 34.5	26	19	**45**

Three findings, in decreasing order of confidence.

The calibration gain saturates early. The first 300 examples buy +9.03 macro-F1; the next 600 buy +0.60. The marginal return falls from 3.0 points per hundred examples to 0.1 — a thirty-fold drop. Whatever mechanism restores the three-class behaviour, it needs very little data to engage. This is the most useful practical result in the project: the fix is cheap.

Cost rises monotonically with budget, and at 300 the main task pays nothing. TextVQA falls −1.23 at 300 and −2.23 at 900. SLAKE closed accuracy is +0.48 at 300 and −1.21 at 900; open exact match rises slightly at both. The 300-example point therefore dominates the 900-example point on cost while capturing 94 % of the macro-F1 gain (92 % once both ends are averaged over two seeds; see the complete curve below) — and the earlier statement that C1's SLAKE cost was "offset" is superseded by the cleaner observation that a smaller budget has no such cost to offset.

The cost is confined to short-answer format. A second retention probe run after the above — MMBench, 500 fixed multiple-choice questions across 20 ability dimensions — moves 0.00 at 300 and +0.60 at 900 relative to zero replay (base 88.40, A-server 87.40, C2-300 87.40, C1 88.00; one point is five questions; zero unparsed answers). The largest per-category movements are single questions. So the TextVQA decline is not a loss of visual ability: it is a perturbation of the short-answer output distribution by the one-word replay targets, which a letter-choice probe does not see. This is consistent with the per-question finding in Section 5.5 that about half of the TextVQA regressions are surface variants. The cost side of the trade-off is therefore smaller than the TextVQA number alone suggests.

But the small budget over-corrects, and the larger one is better balanced. At 300 examples the model swings past the target: "no"→"yes" errors drop from 63 to 14 — below the untuned base model's 42 — but "yes"→"no" errors rise from 18 to 32, "yes" recall falls from 91.7 to 69.9, and the model now predicts 228 "yes" against 276 gold. At 900 the per-class recalls are far more even (79.7 / 70.4 / 34.5 versus 69.9 / 78.7 / 38.2), and the two dangerous error types sum to almost the same total (45 versus 46) but are distributed differently. Which operating point is preferable is therefore a deployment question — minimum general-ability cost favours 300, balanced per-class behaviour favours 900 — and the report declines to name an "optimum" that the data does not single out.

Why the 1,800-example point was not run. The gain has already saturated, so that point would most likely only add cost; and with 55 unique "maybe" items it would repeat each nearly 11 times, so a regression there could not be separated from repetition-driven over-fitting. The more informative missing measurements are a point

between 0 and 300, where all the gain occurs, and a second seed at 300 — the 27-question swing in "yes" recall between 300 and 900 is exactly the size of effect a single run cannot distinguish from noise. The second seed follows.

A second seed at 300 reproduces the point. C2-300 was repeated with both the sampling seed and the training seed changed to 43 — a treatment-level replication that covers sampling and training variance together, so a difference could not be attributed to one alone. It landed on top of the first run:

300 replay	macro-F1	"maybe" correct	"no"→"yes"	"yes"→"no"	yes recall	SLAKE closed	TextVQA	MMBench
seed 42	60.48	21 / 55	14	32	69.9	85.82	82.33	87.40
seed 43	59.80	18 / 55	12	30	72.1	85.82	82.56	87.00
900 (C1)	61.09	19 / 55	26	19	79.7	84.13	81.33	88.00

Run-to-run spread at this scale is therefore about 0.7 macro-F1, about 2 points of single-class recall, and about 2 questions on each dangerous-error count. Against that yardstick the two claims above resolve differently. The saturation claim holds in a tightened form: the 300-example mean (60.14) sits 0.95 below the 900-example run, which is the same order as the seed spread, so most of the gain is in the first 300 examples and the further 600 buy an amount indistinguishable from noise. The over-correction claim is strengthened: "yes" recall is 69.9 and 72.1 at 300 against 79.7 at 900 — an 8.7-point gap, four times the seed spread — and both dangerous-error counts move in the same direction in both seeds ("yes"→"no" 32 / 30 versus 19; "no"→"yes" 14 / 12 versus 26). The swing toward "no" at a small budget is a property of the budget, not of a run.

The complete curve. Two further runs close the design: a 99-example point (33 per class — each "maybe" item appears at most once, so this point also tests whether the fix depends on repeating the 55 unique "maybe" items) and a second seed at 900. The curve is now 0 / 99 / 300 / 900 with two seeds at each end.

Replay	macro-F1	acc	"maybe" correct	predicted yes / no / maybe	yes recall	"no"→"yes"	"yes"→"no"	SLAKE closed	open EM	TextVQA	MMBench
0	51.45	68.80	6 / 55	356 / 112 / 32	91.7	63	18	85.34	75.50	83.56	87.40
99	57.44	64.60	**23 / 55**	197 / 183 / 120	**60.9**	**12**	36	85.34	76.43	82.78	87.20
300, seed 42	60.48	69.40	21 / 55	228 / 178 / 94	69.9	14	32	**85.82**	75.97	82.33	87.40
300, seed 43	59.80	69.80	18 / 55	234 / 176 / 90	72.1	12	30	**85.82**	77.36	82.56	87.00
900, seed 42 (C1)	**61.09**	71.60	19 / 55	269 / 151 / 80	79.7	26	19	84.13	76.12	81.33	88.00
900, seed 43	60.72	**74.40**	12 / 55	287 / 162 / 51	84.4	25	21	84.62	77.21	82.33	87.60

The second seed at 900 widens the noise yardstick: the two 900 runs differ by 0.37 macro-F1 but by 7 "maybe" questions, 4.7 points of "yes" recall and 1.0 point of TextVQA (three questions). Run-to-run spread at this scale is therefore under 0.7 macro-F1, up to about 5 points of single-class recall, up to 7 questions on the "maybe" count, about 2 on each dangerous-error count and up to 1 point on TextVQA. Every claim below is judged against that wider yardstick.

Saturation holds, and the numbers tighten. Relative to zero replay, 99 examples buy +5.99 macro-F1, 300 buy +8.69 (seed mean) and 900 buy +9.46 (seed mean): 63 % and 92 % of the final gain. The marginal return per hundred examples falls 6.0 → 1.3 → 0.13, a fifty-fold decline. The step from 99 to 300 (+2.70) is four times the

seed spread and is real; the step from 300 to 900 (+0.77) is the same size as the spread at either end and cannot be distinguished from noise.

Over-correction is monotone in budget across three budgets and four runs. "Yes" recall is 91.7 with no replay, then 60.9 / 71.0 / 82.1 at 99 / 300 / 900 (seed means); "yes"→"no" errors are 18, then 36 / 31 / 20; "no"→"yes" errors are 63, then 12 / 13 / 25.5. Any amount of replay swings the model past the target, and more replay brings it back toward the gold proportions. The smallest 300-versus-900 gap in "yes" recall (72.1 versus 79.7) exceeds the larger seed spread (4.7). The total of the two dangerous error types is flat across every replay budget (48, 46, 42, 45, 46 against 81 with none): replay fixes the total at once, and the budget only decides how it is split.

The fix does not depend on repetition, but a small budget fixes the prior rather than the judgement. At 99 examples each "maybe" item is seen at most once, yet "maybe" recall (41.8 %) is the highest of any run — so restoring the class does not rest on the 5.5× repetition in C1's sample, which bounds the memorisation concern raised in Section 7. What that point does not have is discrimination: it predicts "maybe" 120 times against 55 gold and "yes" 197 times against 276, and its plain accuracy (64.6) is below the zero-replay run (68.8). Accuracy rises monotonically with budget (64.6 → 69.6 → 73.0) and so does "maybe" precision (19.2 → 21.2 → 23.7). A consistent reading is that a few dozen three-class examples shift the class prior, and only the larger budgets teach when each class applies; this is an interpretation, not a demonstrated mechanism.

On cost, the end points are clear and the middle is inside the noise. TextVQA seed means fall −0.78 / −1.11 / −1.73 relative to zero replay; adjacent points differ by 0.3–0.6, less than the 1.0-point spread between the two 900 runs, so the earlier "monotonic" wording is weakened to a trend: every replay point is below zero replay, and the 900 end is about five questions below. SLAKE closed accuracy at 900 is lower than every other run in both seeds (84.13 / 84.62 against ≥ 85.34), a stable main-task cost of about one point (four to seven questions) that the smaller budgets do not pay; open exact match is at or above the zero-replay value at every replay point. MMBench (87.40 / 87.20 / 87.20 / 87.80, seed means) stays within ±0.6 at all four points, so the conclusion that the letter-choice probe sees no replay cost survives two more points.

The recommendation is unchanged: 300 examples for nine-tenths of the calibration gain at no main-task cost and about three TextVQA questions; 900 for the most even per-class behaviour and the highest plain accuracy, at about one closed-question point and five TextVQA questions.

5.7 Scaling check: the replay experiment on a 35B mixture-of-experts model

Every result so far comes from one 3B model. To test whether the mechanism is a property of small models, the two conditions that carry the argument — SFT with no replay and SFT with 300 replay examples — were repeated on **Intern-S2-Preview** [16], a 35B-parameter vision-language model continued-pretrained from Qwen3.5 [37] (sparse mixture of experts [40], 256 routed experts with 8 active per token, about 3B active parameters; 40 layers of which one in four is full attention and the rest gated linear attention [46]; a 27-layer vision encoder). Both conditions were run with two seeds; as at 3B, the second seed changes the replay sample and the training seed together.

Setup, and what differs from 3B. Training data files, prompts, learning rate, epochs, effective batch 16 (4 × 4), cutoff, image cap and LoRA rank 16 / alpha 32 / dropout 0.05 are identical to the 3B runs. Three things differ, all forced by the architecture:

- *Training code.* LLaMA-Factory does not support the model, so training uses a short transformers + peft script (`train/interns2/train_lora.py`) with the model split layer-wise across four RTX 5090s. Labels cover only the answer and the end-of-turn token, as in LLaMA-Factory.
- *LoRA targets.* The 256 routed experts are stored as packed 3D parameters rather than linear layers, so LoRA is attached to the attention projections (full-attention q/k/v/o, linear-attention qkv/z/out) and the shared

expert only: 250 layers, 19.2M trainable parameters, 0.054 % of the model (3B: all linear layers, 0.79 %). The vision encoder, router, routed experts and embeddings are frozen.

- *Thinking mode.* The model reasons before answering by default; with the evaluation's 8-32 token budget the reasoning consumes the whole answer. Both training and evaluation use the chat template with thinking disabled. Evaluation goes through an OpenAI-compatible LMDeploy server [31] (`medvlm/remote.py`) after the adapter is merged into the weights, so the base and every fine-tuned model are scored through the same serving path.

	3B base	3B replay 0	3B replay 300 (2 seeds)	35B base	35B replay 0 (2 seeds)	35B replay 300 (2 seeds)
SLAKE closed	66.35	85.34	85.82 / 85.82	84.38	94.47 / 93.75	94.47 / 93.99
SLAKE open EM	40.93	75.50	75.97 / 77.36	67.13	86.36 / 86.98	86.36 / 87.44
TextVQA	84.22	83.56	82.33 / 82.56	88.78	88.22 / 86.89	86.67 / 86.89
MMBench	88.40	87.40	87.40 / 87.00	93.60	94.40 / 93.40	94.00 / 94.00
PubMedQA macro-F1	48.66	51.45	60.48 / 59.80	61.83	59.51 / 63.23	**67.29 / 65.38**
PubMedQA acc	64.00	68.80	69.40 / 69.80	79.00	79.00 / 78.60	77.60 / 78.60
predicted "maybe" / 55 gold	83	32	94 / 90	27	18 / 34	75 / 55
"maybe" correct / 55	7	6	21 / 18	7	4 / 10	22 / 15
"no"→"yes"	42	63	14 / 12	20	24 / 22	13 / 9
"yes"→"no"	14	18	32 / 30	17	16 / 16	13 / 18

PubMedQA figures are on the held-out 500; the 3B "replay 0" column is the single A-server run. At 35B the two seeds of one condition differ by up to 3.7 macro-F1 (zero replay) and 7 correct "maybe" answers (replay 300), against about 0.7 macro-F1 at 3B, so single-run differences under about two points are not interpreted at 35B.

The zero-shot 35B model has the same blind spot. It separates "yes" from "no" far better than the 3B base ("no"→"yes" 20 against 42, accuracy 79.0 against 64.0), and its closed SLAKE accuracy without any fine-tuning (84.4) is already level with the fine-tuned 3B model. But it predicts "maybe" 27 times against 55 gold and gets 7 right — exactly the 3B base's count. Scale fixes the binary judgement, not the willingness to abstain.

Short-answer SFT does not reliably push the 35B model toward "yes". This is where the scales differ. At 3B, SFT raised "no"→"yes" errors from 42 to 63 and collapsed "maybe"; at 35B the two zero-replay seeds give "no"→"yes" 24 and 22 against 20, predicted "maybe" 18 and 34 against 27, and macro-F1 59.5 and 63.2 against 61.8 — one seed below the base, one above. The first zero-replay seed alone suggested a weaker version of the 3B effect; the second does not support it. At 35B the calibration problem is the base model's own reluctance to answer "maybe", not a bias the SFT data introduces.

The same 300 examples still repair it, without the 3B over-correction. Both replay seeds score above both zero-replay seeds (67.3 and 65.4 against 59.5 and 63.2; means 66.3 against 61.4, +5.0; 3B: +8.7), answer 22 and 15 "maybe" questions correctly against 4 and 10, and cut "no"→"yes" from 23 to 11 on average. The difference from 3B is in the side effects. At 3B the fix overshoots — "yes"→"no" errors rise to 32 and 30 and the model predicts only 228 and 234 "yes" against 276 gold. At 35B "yes"→"no" stays at 13 and 18, level with zero replay (16) and the base (17), and predicted "yes" is 261 and 271. The larger model absorbs the three-class examples without being pushed past the target. Plain accuracy stays level (77.6 and 78.6 against 79.0 at zero replay) because some of the added "maybe" answers are wrong; macro-F1, which weights the classes equally, is the better summary here.

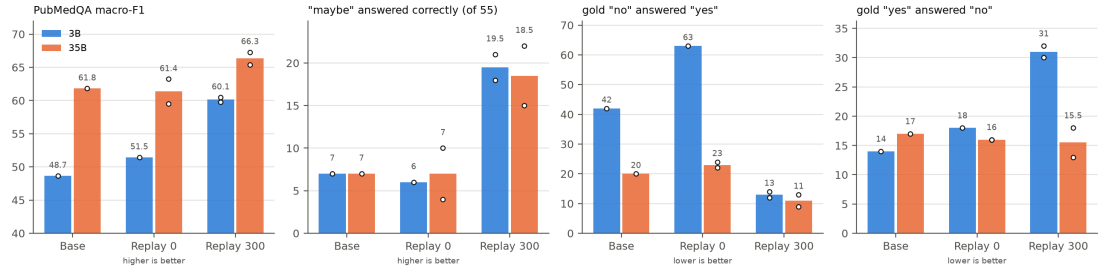


Figure 7. Held-out PubMedQA (500 questions) at both scales, three conditions. Bars are seed means, dots individual runs (the 3B zero-replay condition is a single run). Replay repairs "maybe" at both scales; zero-replay SFT raises "no"→"yes" errors only at 3B, and only at 3B does replay push "yes"→"no" errors up. `scripts/make_fig7_scale.py`.

The main task is unaffected and the retention cost is within noise. Replay leaves SLAKE essentially unchanged (closed 94.23 against 94.11, open exact match 86.90 against 86.67, seed means); SFT itself adds about 10 closed and 20 open points, ending 8-11 points above the fine-tuned 3B model. TextVQA falls from 88.78 at the base to 87.56 with SFT and 86.78 with replay (means), but the two zero-replay seeds alone differ by 1.3 points, so the 0.8-point replay cost cannot be separated from run-to-run spread; MMBench is flat (93.90 against 94.00). The 3B finding that the cost is confined to short-answer format becomes, at 35B, a cost too small to measure.

Group-D ablations on the 35B replay-300 condition. Each run changes one training setting. The control, the attention-only run and the one-epoch run have two seeds; the others are single runs.

Run	Change	macro-F1 (seed 42 / 43)	"maybe" correct	"no"→"yes" / "yes"→"no"	SLAKE closed	open EM	TextVQA
control	rank 16, lr 1e-4, 3 epochs, attention + shared expert	**67.29 / 65.38**	22 / 15	13 / 13 · 9 / 18	94.47 / 93.99	86.36 / 87.44	86.67 / 86.89
attention only	no LoRA on the shared expert (130 layers, 0.040 %)	67.29 / 64.25	25 / 16	10 / 19 · 13 / 23	94.47 / 93.51	85.74 / 86.67	86.00 / 87.00
1 epoch	3 → 1 epoch	64.14 / 62.32	22 / 10	9 / 16 · 13 / 19	93.03 / 93.03	82.79 / 84.34	87.44 / 87.67
lr 5e-5	learning rate halved	64.56	21	13 / 18	92.55	85.89	86.67
lr 2e-4	learning rate doubled	66.28	22	14 / 12	95.19	86.20	86.11
rank 8	rank 8, alpha 16	66.28	23	15 / 15	93.27	86.98	87.44
rank 32	rank 32, alpha 64	66.31	24	17 / 12	93.27	86.51	87.11
reference: replay 0	no three-class examples	59.51 / 63.23	4 / 10	24 / 16 · 22 / 16	94.47 / 93.75	86.36 / 86.98	88.22 / 86.89

The data mix outweighs the training settings. Replay moves macro-F1 by about five points on average; among the training settings only under-training moves it reliably, by about three, and the rest stay within the 35B spread. *Under-training is the only reliably harmful direction.* Both one-epoch seeds fall below both control seeds (mean 63.2 against 66.3), score 93.03 closed accuracy both times and lose about 3.3 points of open exact match; the single halved-learning-rate run points the same way. The first one-epoch seed over-predicted "maybe" (96 times) and the second under-predicted it (46), so the only defensible description is that the class is unstable when training is short. *More training and more capacity do not help:* a doubled learning rate and ranks 8 and 32 land within a point of each other and of the control. *No detectable contribution from LoRA on the shared expert:* attention-only LoRA averages 65.8 over two seeds against the control's 66.3, a third of the seed spread, with closed accuracy one question apart on average; the exact tie of the first seeds was a coincidence. At this noise level the shared-expert adapters, a quarter of the trainable parameters, are not needed.

Replay budget at 35B. To test whether the 3B budget curve (Section 5.6) holds at scale, the 99- and 900-example conditions were run on the 35B model with two seeds each, from the same data files as at 3B (regenerated on the training host with the same script and seeds, and checked byte-for-byte against the existing 300-example file).

Replay	macro-F1 (seed 42 / 43)	mean	accuracy	predicted "maybe" / 55	"maybe" correct	"no"→"yes" · "yes"→"no"	SLAKE closed	open EM	TextVQA	MMBench
0	59.51 / 63.23	61.37	79.0 / 78.6	18 / 34	4 / 10	24 · 16 / 22 · 16	94.47 / 93.75	86.36 / 86.98	88.22 / 86.89	94.40 / 93.40
99	65.58 / 64.64	65.11	74.0 / 72.4	99 / 118	27 / 28	13 · 17 / 8 · 13	93.99 / 93.99	85.74 / 87.13	87.67 / 87.56	93.60 / 93.60
300	**67.29 / 65.38**	**66.34**	77.6 / 78.6	75 / 55	22 / 15	13 · 13 / 9 · 18	94.47 / 93.99	86.36 / 87.44	86.67 / 86.89	94.00 / 94.00
900	64.17 / 65.30	64.74	77.0 / 77.0	55 / 69	15 / 18	21 · 14 / 15 · 12	93.99 / 93.75	86.82 / 86.20	87.22 / 87.33	93.20 / 93.80

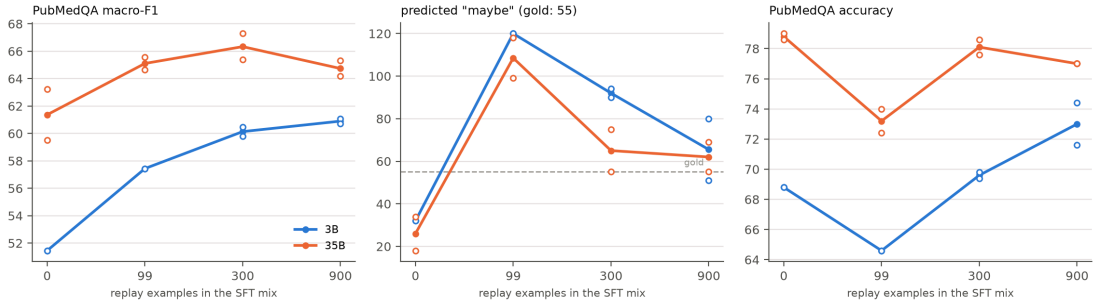


Figure 8. Held-out PubMedQA against replay budget at both scales. Lines join seed means, open dots are individual runs (at 3B the 0 and 99 points are single runs); the budget axis is categorical. `scripts/make_fig8_dose.py`.

Every budget helps, and the curve is flat from 99 to 900. All six replay runs (64.2–67.3) score above both zero-replay runs (59.5 and 63.2). The seed means are 65.1, 66.3 and 64.7 at 99, 300 and 900; 300 is nominally the best, but the steps between adjacent budgets (+1.2, −1.6) are smaller than the spread between the two 300-example seeds (1.9). At 3B the step from 99 to 300 was four times the seed spread; at 35B, 99 examples already buy three-quarters of the best gain (+3.7 of +5.0).

The small-budget over-correction reproduces, in a different direction. At 99 examples both scales predict "maybe" about twice as often as it occurs (99 and 118 at 35B, 120 at 3B, against 55 gold), answer the most "maybe" questions correctly (27 and 28 at 35B) and lose plain accuracy (35B: 74.0 and 72.4, against 78.8 with zero replay and at least 77.0 at every larger budget). At 3B the excess came with doubled "yes"→"no" errors (18 → 36); at 35B "yes"→"no" stays at 17 and 13 (zero replay: 16), so the larger model hedges with "maybe" instead of flipping to "no". From 300 examples on, predicted "maybe" is near the gold count (75 / 55 at 300, 55 / 69 at 900) and accuracy returns to within about two points of zero replay. As at 3B, the largest budget lets "no"→"yes" errors drift back up (21 and 15 at 900 against 13 and 9 at 300), part-way toward the zero-replay level (24 and 22).

No retention cost at any budget. Seed means at 0 / 99 / 300 / 900 examples: SLAKE closed 94.11 / 93.99 / 94.23 / 93.87, open exact match 86.67 / 86.44 / 86.90 / 86.51, TextVQA 87.56 / 87.62 / 86.78 / 87.28, MMBench 93.90 / 93.60 / 94.00 / 93.50. None trends with budget, and the one-point SLAKE cost that 900 examples paid at 3B does not appear. The recommendation of 300 examples therefore holds at both scales — nominally the best macro-F1 at 35B, predicted "maybe" near the gold count and accuracy at the zero-replay level — but at 35B the budget is a forgiving setting: anything from 99 to 900 gives most of the gain, provided it is not so small that the model over-hedges.

Thinking mode does not substitute for replay. Every 35B number above was measured with thinking disabled. To test whether reasoning alone restores "maybe", the zero-shot base was evaluated on the held-out 500 with thinking enabled and a 4,096-token budget (`MEDVLM_MAX_TOKENS`; the answer is read from the text after the reasoning).

Base model, held-out PubMedQA	questions	macro-F1	accuracy	predicted "maybe"	"maybe" correct	"no"→"yes" · "yes"→"no"
thinking off	500	61.83	79.0	27	7 / 55	20 · 17
thinking on, unfinished counted wrong	500	56.28	65.6	20	5 / 55	11 · 9
thinking on, the 395 it finished	395	64.29	83.0	20	5 / 37	11 · 9
thinking off, same 395	395	62.47	83.3	12	3 / 37	14 · 9
replay 300 (thinking off, seeds 42 / 43), same 395	395	67.55 / 67.18	80.8 / 82.8	46 / 31	12 / 9 of 37	–

105 of the 500 answers (21 %) never finished: the reasoning ran past 4,096 tokens without committing to an answer, typically alternating between two labels. Unfinished questions are concentrated where the truth is uncertain — 18 of the 55 "maybe" questions (33 %) against 17 % of "yes" and 24 % of "no" — and the median reasoning length is 2,842 tokens on "maybe" questions against 984 and 1,278 on "yes" and "no". The model's reasoning registers the uncertainty but does not convert it into the answer "maybe". On the 395 questions it did finish, thinking changes 13 answers (4 fixed, 5 broken) and adds 1.8 macro-F1 over the same model without thinking, against 4.7-5.1 for the replay models on the same questions — a comparison that favours thinking, since it drops the questions thinking found hardest. It also costs about 1,200 generated tokens per answer (2 h 49 min for 500 questions). This is one run of the base model at one token budget; a larger budget would finish more questions, but even if every unfinished "maybe" question were then answered correctly, thinking would reach 23 of 55: level with the best 300-example run (22) and below both 99-example runs (27, 28), not beyond them.

Open-ended generation. The three short-output retention probes cannot see drift in longer text, so a fourth probe (`eval/eval_openended.py`) asks for free descriptions. General: 300 fixed COCO val images [25], "describe this image in two or three sentences", scored against five human captions [7] (CIDEr [44] without the CIDEr-D length penalty, which zeroes out multi-sentence answers against ten-word references; ROUGE-L [26]; recall of reference content words) plus length, share of answers under five words and refusal rate. Medical: all 96 SLAKE test images, "describe the imaging modality, the body region and any abnormal findings", scored against the annotated modality and region and, for the 44 images whose own closed questions mark them abnormal, whether the description declares the study normal once negated findings ("no evidence of consolidation or nodules") are removed. All thirteen 35B models were evaluated.

Model	COCO CIDEr	content recall	words	< 5 words	region correct	abnormal called normal
35B base	65.79	71.40	48.2	0 %	96.9	4 / 44
replay 0, seeds 42 / 43	68.18 / 66.52	71.54 / 71.60	46.0 / 46.1	0 %	94.8 / 95.8	**13 / 3**
replay 300, seeds 42 / 43	68.84 / 68.01	72.54 / 71.42	44.5 / 45.3	0 %	94.8 / 93.8	2 / 4
eight other ablation runs	61.5 – 69.8	71.4 – 73.0	43.8 – 52.3	0 %	89.6 – 95.8	2 – 15

Long-form general description does not degrade: all twelve fine-tuned models write 44-52 words (base 48), none collapses to a short answer or refuses, and caption similarity is unchanged. The medical false-normal count is a cautionary result. The first zero-replay seed described 13 of the 44 abnormal studies as normal against 4 for the base and 2 for replay 300 — all 13 were real misses on inspection, including two chest films where the base and replay models reported an enlarged cardiac silhouette and the zero-replay model wrote "the heart size appears normal". The second zero-replay seed produced 3. Across the twelve fine-tuned models the count ranges from 2 to 15 (a replay model at rank 8 has 15) and is unrelated to PubMedQA calibration (Spearman −0.27). A single decoding over 44 images is too coarse a measure: the earlier single-seed reading that replay reduces false-normal descriptions is withdrawn. Turning missed abnormalities into a usable metric needs a larger annotated set, several seeds and human reading.

5.8 External test sets

The tables so far measure SLAKE, which is also the training set, and three general probes. After the week-3 review the instructor approved three English public test sets proposed for this purpose, and a fourth was taken from data shared by a classmate. All are evaluation-only; only their test shards are downloaded (Data Card, Section 4d). **VQA-RAD** [23] (451 radiology questions, 251 yes/no) asks whether medical fine-tuning transfers to a different source. **PathVQA** [14] (6,719 pathology questions, 3,362 yes/no) asks whether the yes/no bias of Sections 5.3–5.7 appears on image questions in another domain. **MedQA** [19] (1,273 four-option USMLE questions) asks whether text-only medical knowledge is kept. **PneumoniaMNIST** (624 paediatric chest X-rays, 390 pneumonia / 234 normal; MedMNIST test split [47]) is asked both "does this X-ray show pneumonia? yes or no" and for a free description, to measure missed abnormalities on a set ten times larger than the 44 SLAKE images of Section 5.7. Prompts and scoring for the VQA sets are those of SLAKE; MedQA is scored like MMBench. The five 35B models are the base and replay 0 / replay 300 with two seeds each.

	base	replay 0 (s42 / s43)	replay 300 (s42 / s43)
VQA-RAD closed	80.48	85.26 / 84.06	85.26 / 83.27
VQA-RAD open EM	33.0	34.5 / 35.5	35.0 / 34.5
PathVQA closed	80.13	76.23 / 77.57	76.83 / 77.84
PathVQA predicted "yes" (gold 54.0 %)	56.7 %	46.4 / 50.4 %	49.6 / 51.3 %
PathVQA "yes"→"no" / "no"→"yes"	288 / 379	524 / 269 · 433 / 315	458 / 312 · 412 / 324
MedQA	85.00	83.97 / 83.42	83.58 / 83.90
PneumoniaMNIST yes/no accuracy	84.94	83.01 / 84.78	84.29 / 81.41
PneumoniaMNIST sensitivity / specificity	80.5 / 92.3	78.5 / 90.6 · 91.0 / 74.4	82.3 / 87.6 · 77.7 / 87.6
pneumonia films described without a lung abnormality	67.2 %	74.9 / 41.5 %	77.7 / 70.0 %
of those answered "yes, pneumonia": described without one	59.5 %	68.0 / 35.8 %	73.2 / 61.4 %

Fine-tuning transfers to another radiology set. All four fine-tuned runs gain 2.8–4.8 closed-question points on VQA-RAD (7–12 of 251 questions); open exact match moves by at most 2.5. The SLAKE gains are therefore not specific to the SLAKE question distribution.

On pathology the same fine-tuning costs accuracy and tilts the yes/no balance the other way. All four runs lose 2.3–3.9 closed points on PathVQA. The base answers "yes" slightly more often than the gold (56.7 % against 54.0 %); every fine-tuned run answers it less often (46.4–51.3 %), and "yes"→"no" errors rise from 288 to 412–524. The shift is a property of fine-tuning, present with and without replay. At 3B, short-answer fine-tuning pushed PubMedQA toward "yes"; here it pushes pathology toward "no". What the SLAKE data instils is a domain-specific answer prior, not a fixed direction of bias, which is consistent with the calibration argument of Section 6.

A small, consistent loss of text knowledge. MedQA falls by 1.0–1.6 points (13–20 of 1,273 questions) in every run, mostly on step-2/3 questions. It is the only general probe at 35B whose cost is consistent across seeds (Section 5.7 found TextVQA and MMBench within noise).

Replay makes no measurable difference on the external sets. On every external measure the gap between replay 0 and replay 300 is smaller than the gap between two seeds of one condition (for example 4.0 points of predicted-"yes" rate on PathVQA between the zero-replay seeds). A single-seed reading that replay softens the pathology shift was withdrawn when the second seeds were run.

PneumoniaMNIST, yes/no question: unchanged accuracy, unstable threshold. Accuracy is 81–85 for all five models, but the decision threshold moves from run to run: one zero-replay seed answers "yes" to 66.5 % of films (35 missed pneumonias, 60 false alarms) while one replay seed misses 87. Missed-pneumonia counts are not comparable between conditions.

PneumoniaMNIST, free description: the measure needed fixing, and then shows a stable inconsistency rather than a training effect. The first count reused the "claims normality" judge of Section 5.7. A hand check found it misses phrasings such as "no apparent abnormalities" and "absence of major pulmonary abnormalities", and that it misses different numbers of descriptions for different models (21–99 of 624), which distorts comparisons; an early reading that every fine-tuned model separates pneumonia from normal films better than the base was withdrawn on that basis. The judge was replaced by the opposite question — after negated phrases are removed, does the description still report a lung abnormality — and a hand check of 50 descriptions across the five models found all 50 judged correctly (`scripts/rescore_pneumonia.py` recomputes it from stored descriptions). Under this judge the base describes 67 % of pneumonia films without any lung abnormality and the fine-tuned runs 42–78 % — three runs worse than the base, one much better — so training conditions cannot be ranked. What is stable across all five models is an inconsistency between the two question formats: of the films a model has just called pneumonia when asked yes or no, 36–73 % (base 60 %) are described without any lung abnormality. The model recognises the finding when asked about it and omits it when asked to describe the film — the same pattern as Section 5.7's thinking-mode result, where the reasoning registers uncertainty that the answer does not express. With 390 abnormal images the run-to-run spread is still large, so the instability of free-text abnormality reporting is a property of the fine-tuned models, not only of the 44-image sample used before.

5.9 Experiment B3: image-text CPT on CheXpert Plus, filtered by alignment score

B3 asks whether the quality of image-text CPT data matters, using the image-report alignment model of the Topic 1 teammate as the quality score. The CheXpert Plus subset [4] prepared by the teammate (15,000 frontal pairs, patient-level split) was the planned source, but the alignment model reaches 17–28 % image→text R@1 on its own training pairs against 1.9 % on validation, so scores on those pairs would mostly measure memorisation. The candidate pool was therefore drawn from the rest of CheXpert Plus: 10,000 studies from 10,000 patients outside the teammate's subset (one frontal image per study, one study per patient, report = Findings + Impression), converted with the teammate's own pre-processing code and checked against his image cache (20 of 20 images pixel-identical). The model was loaded read-only in our environment and reproduced his validation retrieval exactly (R@1 0.0191 both ways); matched pairs score 0.30 on average against 0.05 for mismatched ones, and the pool's scores (mean 0.313, quartiles 0.228 / 0.324 / 0.412) match the validation split, so they are not inflated by memorisation.

From the pool, **B3-top** takes the 5,000 highest-scoring pairs (mean 0.421) and **B3-rand** 5,000 pairs drawn at random (mean 0.312); the two sets share 2,481 pairs and are alike in report length (55 and 57 words) and in the share of reports with a Findings section (27 and 30 %; most CheXpert Plus reports carry only an Impression). Each arm runs the B2 recipe unchanged — caption-style CPT for one epoch, then SLAKE SFT — so the arms differ only in how the CPT data were chosen. Images were downscaled once to a 1,024-pixel long side for both arms. The reference is the A-server run (same server, same precision, no CPT).

	A-server (no CPT)	B3-rand	B3-top
SLAKE closed	85.34	85.58	85.58
SLAKE open EM	75.50	75.81	76.74
TextVQA	83.56	82.33	82.00
MMBench	87.40	87.40	88.00
PubMedQA macro-F1 (held-out)	51.45	52.02	50.05
PubMedQA "maybe" correct / "no"→"yes"	6 / 63	6 / 59	3 / 60
CPT loss, mean of last 10 logs	–	1.200	1.176

CheXpert Plus CPT does not help the main task. Both arms land within one closed SLAKE question of the run without CPT and cost 1.2–1.6 TextVQA points (four to five questions), with MMBench unchanged and the

PubMedQA "yes" bias untouched. This is the B1 / B2 result again with a larger, adult, better-matched corpus: with a frozen vision tower and a few thousand caption examples, the CPT stage adds nothing measurable after SFT.

Alignment-score filtering shows no measurable effect. The top arm is 0.9 open exact-match points above the random arm (13 questions better, 7 worse, of 1,061), which is below the 1.4-point seed spread of open exact match at 3B; the other tables differ by less. The one consistent sign is a slightly lower CPT loss on the better-aligned pairs (1.176 against 1.200), which does not carry through SFT. Four reasons limit what this null result can say: one seed per arm; half of the data is shared between arms, so only about 2,500 pairs differ; the scorer itself is weak (1.9 % R@1 on 1,518 candidates); and the downstream SLAKE tables barely respond to CPT at all, so even a real difference in CPT quality has little room to show. A more sensitive test would compare the two CPT adapters directly on report generation for the teammate's held-out CheXpert Plus test split, before any SFT.

5.10 Missed abnormalities on adult chest X-rays (direction A)

Sections 5.7 and 5.8 could not turn missed abnormalities into a usable measure: 44 abnormal SLAKE images were too few, and PneumoniaMNIST is paediatric and low-resolution. The test split of the Topic 1 teammate's CheXpert Plus subset [4] provides 3,189 adult frontal films whose CheXbert labels [17], extracted automatically from the reports, mark 2,733 as abnormal (any positive finding other than support devices), 231 as "No Finding" and 225 as carrying only uncertain labels. It was used read-only, for evaluation only, with the teammate's agreement. Each film was asked "Are there any abnormal findings in this chest X-ray? Answer with yes or no only." and for a free description; the description judge is the negation-aware rule validated in Section 5.8, re-checked here by hand on 30 descriptions of the replay-300 model (15 per verdict, all correct). The data never left the server: the evaluation ran in the container that holds it, calling models served on the GPU host of the same machine.

	base	replay 0 (s42 / s43)	replay 300 (s42 / s43)
yes/no: sensitivity (abnormal films answered "yes")	87.1	96.3 / 98.0	95.2 / 97.6
yes/no: missed abnormal films (of 2,733)	352	100 / 55	130 / 66
yes/no: specificity (normal films answered "no")	81.8	54.1 / 38.1	58.9 / 41.6
yes/no: uncertain-only films answered "yes" (%)	38.2	69.8 / 80.0	62.7 / 76.9
description: abnormal films described with no abnormality (%)	29.2	52.5 / 14.6	68.5 / 56.6
description: normal films described as clean (%)	81.4	91.8 / 66.7	94.8 / 91.8
description: discrimination (points, clean-normal minus clean-abnormal)	52.2	39.3 / 52.0	26.4 / 35.2
films answered "yes" yet described as clean (%)	21.6	50.7 / 12.9	66.9 / 55.5

Asked directly, every fine-tuned model leans toward "abnormal". Sensitivity rises from 87.1 to 95.2–98.0 and missed abnormal films fall from 352 to 55–130, but specificity falls from 81.8 to 38.1–58.9, and films with only uncertain labels are called abnormal 63–80 % of the time instead of 38 %. All four runs move the same way. Together with the 3B "yes" bias on PubMedQA (Sections 5.3–5.6) and the shift toward "no" on PathVQA (Section 5.8), this shows short-answer fine-tuning resetting the yes/no threshold, with a direction that depends on the domain.

Asked to describe, most fine-tuned models omit more abnormalities, and by a seed-dependent amount. The base describes 29.2 % of abnormal films without any abnormality; the fine-tuned runs describe 52.5 and 14.6 % (replay 0) and 68.5 and 56.6 % (replay 300). The ability to separate normal from abnormal films in free text (52.2 points for the base) is never higher after fine-tuning (39.3, 52.0, 26.4, 35.2). Both replay-300 runs are worse than both replay-0 runs on these two measures, but the two replay-0 seeds differ by 38 points, so that ordering is suggestive only. With 2,733 abnormal films the seed dependence cannot be a sample-size artefact: it is a property of the fine-tuned models.

The two question formats contradict each other more. Of the abnormal films a model calls abnormal when asked, the base describes 21.6 % as clean; three of the four fine-tuned runs describe 51–67 % as clean (one, 12.9 %). A model can answer "yes, abnormal" for 98 % of abnormal films and still write a normal-sounding description for most of them.

Specific findings are rarely named, even by the base.

finding named in the description (%)	base	replay 0 (s42 / s43)	replay 300 (s42 / s43)
Pleural Effusion (1,445 films)	33.8	20.5 / 28.4	14.5 / 26.2
Lung Opacity (1,586 films)	70.4	49.5 / 84.7	30.4 / 46.3
Edema (900 films)	14.1	12.7 / 16.1	3.2 / 5.3
Cardiomegaly (405 films)	25.4	25.4 / 38.5	21.0 / 19.8
Pneumothorax (278 films)	9.7	8.3 / 10.4	2.9 / 4.7

Keyword matching here is lenient ("fluid" counts as an effusion) and was not hand-validated, so these rates are upper estimates; a pneumothorax, an emergency, is named in at most 10.4 % of descriptions (base 9.7 %). A hand reading of the 30 checked descriptions also found them formulaic: clean descriptions are near-identical sentences, and abnormal ones repeatedly place "an opacity in the lower right lung field", which suggests a default phrase rather than a reading of the film. This is a qualitative observation, not a measured rate.

Two qualifications apply. The labels are produced by CheXbert from report text, not by radiologists reading the images, and the normal class is small (231 films), so specificity is the least precise number in the table. And each model was decoded once, greedily; the seed dependence above is between training runs, not between decodings.

5.11 A model-based judge for the free-text measure (CheXbert)

The keyword judge of Section 5.10 was validated on only 30 descriptions. Every description of six 35B models was therefore re-read by CheXbert [42], the same report labeller that produced the CheXpert Plus reference labels, run on the CPU of the container that holds the data (about two seconds per sentence; one model takes several hours). A description counts as omitting the abnormality when CheXbert finds no positive finding other than support devices. The judge was first checked on 300 radiologist reports, where it agrees with the dataset's own labels on 96.2 % of them. Snapshot: `results/interns2_chexbert_2026-10-18.json`.

35B	omitted, CheXbert (%)	omitted, keyword (%)	answered "yes", described clean (%)	normal films described as abnormal (%)	agreement of the two judges (%)
base	24.95	29.2	18.1	33.8	94.4
replay 300, s42	56.53	68.5	55.0	19.5	86.9
replay 300, s43	35.86	56.6	34.8	33.3	78.1
replay 0, s42	43.80	52.5	41.8	20.4	91.0
replay 0, s43	9.77	14.6	8.3	50.6	93.8
S2-6datasets (Section 5.12)	11.20	13.8	7.6	35.5	95.6
S2-CheXpertPlus (Section 5.13)	10.20	14.0	4.3	26.8	96.2

The ordering of models is unchanged, so the conclusions of Section 5.10 hold under the stronger judge. The keyword rule over-counts omissions for every model, by 2.6–20.7 points, and most on the most formulaic descriptions (the replay-300 s43 model, where the two judges agree on only 78 % of films). CheXbert is the primary judge from here on, with the keyword rule as a cross-check. The seed dependence also survives: the same recipe gives 43.8 % and 9.8 % (replay 0) and 56.5 % and 35.9 % (replay 300). And the models that omit

least describe more normal films as abnormal (replay 0 s43: 50.6 % of 231 normal films), so a lower omission rate is partly bought with false alarms.

5.12 Six training sets in one SFT mix (S2-6datasets)

At the instructor's request the 35B model was retrained on the training splits of all six evaluation datasets, keeping every setting of the replay-300 model (Section 5.7) and changing only the data: SLAKE 4,919, PubMedQA replay 300, VQA-RAD 734, PathVQA 5,000, MedQA 3,000 and PneumoniaMNIST 2,000, 15,953 examples in all. Test items are excluded by image and by question; 202 of the 203 VQA-RAD test images also appear in its official training split, so the 1,059 training questions on those images were removed (data card §4e). 2,991 steps, 5 h 40 min, one seed. For this model the four external sets of Section 5.8 are in-domain (training splits seen, test items not), so they measure in-domain gain, not transfer.

35B	SLAKE closed	SLAKE open EM	TextVQA	MMBench	PubMedQA macro-F1	"maybe" correct / 55	VQA-RAD closed	PathVQA closed	MedQA	PneumoniaMNIST acc / missed of 390
base	84.38	67.13	88.78	93.6	61.83	7	80.48	80.13	85.00	84.94 / 76
replay 300 (s42)	94.47	86.36	86.67	94.0	67.29	22	85.26	76.83	83.58	84.29 / 69
S2-6datasets	**94.71**	**86.20**	**87.22**	**93.4**	**66.59**	**23**	**83.27**	**92.65**	**85.31**	**95.35 / 4**

On the four main tables the extra 10,734 examples neither help nor hurt: every difference from the replay-300 model is smaller than the gap between its two seeds. *In-domain the gains are large where the domain was missing.* PathVQA closed accuracy rises 12.5 points over the base and 15.8 over replay 300, and the shift toward "no" of Section 5.8 disappears (predicted "yes" 53.1 % against 54.0 % true); PathVQA training and test splits share no images, so this is learning the domain rather than recognising images. PneumoniaMNIST misses 4 of 390 pneumonia films instead of 69–76, and MedQA returns to the base level. VQA-RAD does not move, with only 734 training questions left after decontamination.

Free-text omission falls although no description was trained. PneumoniaMNIST was trained with yes/no questions only, yet pneumonia films described without any lung abnormality fall from 67–78 % to 20.8 %, and films called pneumonia but described as clean from 60–73 % to 19.9 %. On the 3,189 adult CheXpert Plus films of Section 5.10, which this model never trained on, omission falls to 13.8 % by keyword and 11.2 % by CheXbert (replay 300: 68.5 / 56.5 %), self-contradiction to 10.2 %, and discrimination reaches 57.2 points, the first fine-tuned 35B model above the base (52.2). It is a single seed, however, and the replay-0 recipe alone spans 9.8–43.8 % on this measure (Section 5.11), so the adult-film result is provisional until a second seed is trained.

5.13 Report-only fine-tuning on CheXpert Plus (S2-CheXpertPlus)

To test whether report writing itself teaches the model to mention abnormalities, a 35B adapter was trained only on image → report pairs: all 10,000 pairs of the B3 pool (Section 5.9; patients disjoint from the 3,189 test films), one of three report-writing prompts per image, target "Findings + Impression", other settings as in Section 5.7. 1,875 steps, 3 h 19 min, final loss 0.757, one seed. The data were generated in the container and moved to the GPU host over the machine's internal network only.

35B	SLAKE closed / open EM	PubMedQA macro-F1 / "maybe" correct	PathVQA closed	MedQA	adult films: omitted (CheXbert / keyword)	discrimination	contradiction	yes/no missed	false alarm (100 – specificity)
base	84.38 / 67.13	61.83 / 7	80.13	85.00	25.0 / 29.2 %	52.2	21.6 %	12.9 %	18.2 %

35B	SLAKE closed / open EM	PubMedQA macro-F1 / "maybe" correct	PathVQA closed	MedQA	adult films: omitted (CheXbert / keyword)	discrimination	contradiction	yes/no missed	false alarm (100 – specificity)
replay 300 (s42)	94.47 / 86.36	67.29 / 22	76.83	83.58	56.5 / 68.5 %	26.4	66.9 %	4.8 %	41.1 %
S2-6datasets	94.71 / 86.20	66.59 / 23	92.65	85.31	11.2 / 13.8 %	57.2	10.2 %	4.5 %	39.4 %
S2-CheXpertPlus	**83.89 / 64.50**	**60.21 / 4**	**78.61**	**84.45**	**10.2 / 14.0 %**	**59.6**	**7.7 %**	**8.7 %**	**25.1 %**

On adult chest films this is the most balanced 35B model: omission is as low as S2-6datasets, discrimination and contradiction are the best of all runs, and the false-alarm rate (25.1 %) is far below the other fine-tuned models (39–41 %), so the lower omission is not bought by calling everything abnormal. Everything it was not trained on returns to the base level: SLAKE short answers, PubMedQA "maybe" (4 correct), the external sets, while TextVQA (87.33) and MMBench (94.0) do not drop. On paediatric PneumoniaMNIST films, description omission improves only partly (67.2 → 53.6 %, against 20.8 % for S2-6datasets), so what adult reports teach transfers little to low-resolution paediatric films. One seed.

5.14 Three-stage training: knowledge pre-training, instruction tuning, reasoning training

The last experiment tests, at 35B and with a much larger corpus than B1–B3, the full pipeline that Task 1.3 describes: (1) text-only continued pre-training on medical knowledge, (2) instruction tuning, (3) reasoning training. All adapters use the LoRA setting of Section 5.7 and are trained one stage on top of the previous one.

- **Stage 1 (`interns2_cpt1`)**: 29 open medical textbooks from OpenMedText [8] (7.4 M tokens), clinical guidelines from the Meditron corpus [6] (4.0 M) and the Chinese medical encyclopedia of MMedC [36] (8.0 M; 622 promotional entries removed). 19.4 M tokens in 24,043 chunks of up to 1,000 tokens, one epoch. The learning rate was chosen before training by a 120-step sweep (5e-5 / 1e-4 / 2e-4 → held-out loss 1.684 / 1.667 / 1.654) and a fixed rule (the smallest rate within 1 % of the best), giving 1e-4. Held-out loss on 235 chunks falls from 1.735 to 1.596 with no sign of over-fitting.
- **Stage 2**: the 15,953 multimodal examples of S2-6datasets plus 12,000 Chinese medical dialogues, 6,000 each from HuatuoGPT-sft-data-v1 [48] and DISC-Med-SFT [3] (cMedQA-derived items excluded for licence reasons; promotional and malformed dialogues filtered), 27,953 examples, three epochs, lr 1e-4. **A** (`interns2_s2a`) starts from the base model; **B** (`interns2_s2b`) starts from the stage-1 adapter.
- **Stage 3 (C, `interns2_s3c`)**: 13,000 exam questions with worked explanations, starting from B: CMExam 5,000 [27], MedMCQA 5,000 [35] and 1,500 Chinese plus 1,500 English final answers from medical-o1-reasoning-SFT [5]. The target is "answer, then a short explanation" with thinking disabled, matching how the models are served; one epoch, lr 5e-5.

Every training item was screened against every test set (a 20-character window matching more than half of a test item removes it; 211 CMExam training questions duplicated its own test split). Training losses (mean of the first → last 10 % of steps): A 1.409 → 1.255, B 1.399 → 1.252, C 1.253 → 1.202. Two Chinese tests were added: the Chinese half of SLAKE (1,033 questions, 420 closed) and 1,000 CMExam single-choice questions. A no-image control asks the SLAKE questions without the image. Snapshots: `results/interns2_3stage_train_2026-10-18.json` and `interns2_3stage_eval_2026-10-19.json`.

test	base	S2-6datasets	stage 1 only	A (2)	B (1→2)	C (1→2→3)
SLAKE closed	84.38	94.71	87.74	94.23	93.51	93.75
SLAKE open EM	67.13	86.20	64.34	84.96	83.72	82.64

test	base	S2-6datasets	stage 1 only	A (2)	B (1→2)	C (1→2→3)
SLAKE, no image: closed / open	—	—	50.48 / 10.70	53.37 / 28.06	49.52 / 26.82	49.28 / 23.88
SLAKE Chinese: closed / open EM	84.29 / 69.98	—	82.38 / 66.88	93.10 / 76.02	94.05 / 76.35	93.57 / 77.81
CMExam	93.0	—	93.8	93.9	93.2	93.8
PubMedQA macro-F1 / "maybe" correct	61.83 / 7	66.59 / 23	59.82 / 4	65.41 / 24	64.16 / 26	64.33 / 22
TextVQA / MMBench	88.78 / 93.6	87.22 / 93.4	87.11 / 93.2	87.89 / 94.2	88.11 / 92.8	88.11 / 93.4
VQA-RAD closed / PathVQA closed	80.48 / 80.13	83.27 / 92.65	83.27 / 80.01	85.66 / 92.18	83.67 / 91.34	82.87 / 91.02
MedQA / PneumoniaMNIST	85.00 / 84.94	85.31 / 95.35	85.70 / 83.49	85.55 / 93.11	85.94 / 93.27	86.41 / 94.55
PneumoniaMNIST: description omits pneumonia (%)	67.2	20.8	83.9	29.2	25.4	43.1
adult films: description omitted, keyword (%)	29.2	13.8	18.4	19.8	22.4	24.4
adult films: contradiction / false alarm (%)	21.6 / 18.2	10.2 / 39.4	9.7 / 18.6	16.2 / 39.4	18.9 / 38.1	19.8 / 32.9

Differences are tested by paired bootstrap over test items (2,000 resamples, `scripts/bootstrap_ci.py`); a difference counts only when its 95 % interval excludes zero.

The gains come from instruction tuning. The stage-1 adapter alone stays at the base level on image questions (SLAKE open 64.3) and is significantly below A on every image test; A, B and C are all far above the base.

Knowledge pre-training adds no measurable improvement (B − A). At seed 42, SLAKE closed −0.7 [−1.9, +0.2], open −1.2 [−3.1, +0.6], Chinese closed +1.0 [−0.5, +2.4], Chinese open +0.3 [−1.6, +2.3], VQA-RAD −2.0 [−4.8, +0.8], MedQA +0.4 [−0.8, +1.5], CMExam −0.7 [−1.8, +0.4], PneumoniaMNIST +0.2 [−0.8, +1.1]. The one interval that excludes zero is PathVQA closed, −0.8 [−1.5, −0.2], against B. The 35B result therefore agrees with B1, B2 and B3 at 3B: a continued pre-training stage, here with 19 M tokens of textbook, guideline and encyclopedia text, yields no measurable improvement at seed 42; the second-seed comparison below tests whether this conclusion persists.

Reasoning training adds almost nothing (C − B, seed 42 only). Only PneumoniaMNIST moves beyond noise, +1.3 [+0.2, +2.4]; Chinese open answers +1.5 [0.0, +2.9] sit on the boundary, and MedQA (86.4, the highest of all models) is +0.5 [−0.6, +1.6]. MedQA and CMExam are close to their ceiling for this model (85–86 % and 93–94 %), which limits what exam-style training can show.

The models use the image. Without it, closed SLAKE accuracy falls to 49–53 %, the level of guessing, and open exact match to 24–28 %. A smaller no-image score is not worse here: B's no-image closed score is 3.9 points below A's [−7.2, −0.2] at seed 42; this difference does not persist at seed 43 (+0.24 [−5.29, +6.01]).

Chinese open answers are not repaired by Chinese dialogue. Chinese closed questions improve as much as English ones (+9 to +10), but Chinese open exact match stays at 76–78, no better than the replay-300 model trained on English only (76.8) and about 8 points below English. Free-form Chinese dialogue does not teach the short Chinese answer vocabulary the SLAKE references use.

A second SFT seed confirms no reproducible gain from knowledge pre-training. A and B were retrained with seed 43, changing LoRA initialisation and data order while keeping the data and settings fixed; B43 still starts from the seed-42 stage-1 adapter. The table compares within-seed B − A differences, in percentage points, with paired 95 % bootstrap intervals. SLAKE closed is −0.72 at both seeds with intervals crossing zero; PathVQA closed falls by less than one point at both seeds with intervals excluding zero. The seed-43 gains on PathVQA open and PneumoniaMNIST are not supported by intervals excluding zero at seed 42, so they are not reproducible

improvements. Snapshots: `results/interns2_seed43_eval_2026-10-25.json`, `results/interns2_ci_seed43_BvsA_2026-10-25.json` and `results/interns2_ci_seed43_vsA42_2026-10-25.json`.

Test	A42 / B42	B – A, seed 42 [95% CI]	A43 / B43	B – A, seed 43 [95% CI]
SLAKE closed	94.23 / 93.51	−0.72 [−1.92, +0.24]	93.75 / 93.03	−0.72 [−1.92, +0.48]
SLAKE open EM	84.96 / 83.72	−1.24 [−3.10, +0.62]	83.26 / 83.57	+0.31 [−1.24, +1.86]
SLAKE Chinese closed	93.10 / 94.05	+0.95 [−0.48, +2.38]	94.76 / 95.24	+0.48 [−0.95, +1.90]
PathVQA closed	92.18 / 91.34	−0.83 [−1.46, −0.21]	92.24 / 91.58	−0.65 [−1.22, −0.06]
PathVQA open EM	31.28 / 31.67	+0.39 [−0.33, +1.07]	30.56 / 31.67	+1.10 [+0.39, +1.88]
PneumoniaMNIST closed	93.11 / 93.27	+0.16 [−0.80, +1.12]	94.39 / 95.67	+1.28 [+0.32, +2.40]
MedQA	85.55 / 85.94	+0.39 [−0.79, +1.49]	86.41 / 85.86	−0.55 [−1.81, +0.71]
Adult-film omission, keyword (%) ↓	19.76 / 22.36	+2.60; CI not computed (per-image labels stay on the server)	15.84 / 18.55	+2.71; CI not computed (per-image labels stay on the server)

Table notes. Scores are percentages; differences and intervals come directly from the paired-bootstrap snapshots, so a difference may differ by 0.01 from subtracting the displayed rounded scores. Adult-film omission is keyword-judged and lower is better; CheXbert re-reading of both seeds is pending. Its per-image labels stay on the server, and no paired interval was computed for this measure.

The effect of Chinese dialogue on adult chest-film omission remains unresolved. A43 omits abnormalities on 15.84 % of films versus A42's 19.76 %, a change of about four points from the seed alone. It lies between the pre-specified outcomes of returning to 11–14 % or remaining around 20 %; seed variation therefore accounts for at least part of the gap from S2-6datasets, and the data do not establish that Chinese dialogue worsens descriptions. B has about 2.7 points more omissions than A at both seeds, but these keyword results await CheXbert re-reading of A42, B42, A43, B43, C and the stage-1 adapter before drawing conclusions about adult-film descriptions.

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6. Analysis

Where SFT gains come from. The largest jumps after SFT are on Organ (+61.6 open), Knowledge-graph (+50.5) and the closed Modality/Plane/Colour/Size types (to 100 %). These are questions the base model could already perceive but answered with the wrong vocabulary or format ("the lungs" versus "lung", full sentences versus a phrase). SFT is largely *vocabulary and format alignment* here. Genuine perception questions moved less: Abnormality-open reaches only 41.5 % and Position-open 57.7 %. Remaining errors are left/right and upper/lower confusions and lesion-category mix-ups.

MRI: data, not capacity. MRI was the weakest modality at baseline (56.8) and becomes the strongest after SFT (93.2). SLAKE's training set is MRI-heavy, so the base model lacked domain exposure rather than the ability to read MR images.

Image-text CPT does not help either, and slightly hurts lesion questions. B2 matches A on closed questions (84.1 versus 85.1) and is 0.6 below on open recall; per question it fixes 11 and breaks 18. The one type that moves beyond noise is Abnormality-open, down from 41.5 to 31.7 (four of 41 questions). This is consistent with Section 5.3: the CPT stage instilled a "normal chest" prior in the language model without changing the visual

features, and that prior is a liability on lesion questions. The X-ray closed-question gain (+0.9, one question of 114) is within noise. Taken together, B1 and B2 say that in this regime, a 3B model with a frozen vision tower and a few thousand CPT examples, the CPT stage contributes nothing to the primary metric; text CPT helps only text tasks. Two changes would be needed for a CPT stage to matter: unfreezing the vision tower (or LoRA on the ViT), or a label-balanced image-report corpus (CheXpert Plus sampled by finding). The report therefore treats A as the main result and B1/B2 as controlled evidence on when CPT does not help.

Text CPT does not cross the modality gap. B1 and A are indistinguishable on SLAKE: every metric within ±0.6, per-question changes 15 fixed versus 14 broken. Two explanations, both plausible: the CPT corpus (abstract-level biomedical text) does not contain the localisation and lesion vocabulary SLAKE tests, and CPT updates only the LLM while the vision side is frozen. The CPT stage is nevertheless effective in its own domain: PubMedQA improves by 2.2 accuracy and 1.8 macro-F1. This motivates image-text CPT (B2) as the only CPT variant that can plausibly help the primary metric.

Forgetting is invisible for SFT alone and appears once text-only data is mixed in. Medical SFT on its own does not move the retention probe: TextVQA is 83.89 / 83.89 / 83.78 for base / A / B1 on Kaggle, and 84.22 → 83.56 for base → A-server on the server. Of the 300 answers, 81 differ from the base only in capitalisation. LoRA with a frozen vision tower and three epochs on 4.9k examples is simply a mild intervention. C1 is the first run to move the probe (−2.23 against A-server), and the control in Section 5.5 attributes that to the replay data rather than to instruction tuning. This is a real signal on a blunt instrument: 2.23 points is about seven questions of 300, and the probe's short-answer OCR format is close to the SFT output format, so the obvious worry was that it under-reports drift in longer-form ability — which made a second, format-insensitive benchmark the main measurement gap in the project until Section 5.6. The second probe resolves that worry in an unexpected direction: MMBench (500 multiple-choice items, `eval/eval_mmbench.py`, now a fourth table in `train/eval_all.sh`) shows the opposite: the replay cost that TextVQA measures at −1.23 / −2.23 is 0.00 / +0.60 on MMBench, within noise. TextVQA *over*-reports relative to a format-insensitive probe, because what replay perturbs is the short-answer output distribution. Two probes with different formats now agree that instruction tuning itself costs about one point (MMBench −1.00, TextVQA −0.66) and that replay costs nothing beyond that except on short-answer tasks.

Reliability worsens in a specific way, the SFT stage is the cause, and the SFT data mix is also the fix. Section 5.3 shows the CPT-only adapters are the best calibrated of all models; the short-answer SFT stage is what creates the bias. Every SFT-containing stage increases willingness to commit: predicted "maybe" falls from 158 (base) to 59 (A) to 44 (B1) against 110 gold. Accuracy rises because SLAKE-style short-answer training removes hedging, not because calibration improves.

Experiment C1 (Section 5.5) closes this loop. If the bias is introduced by the answer format of the SFT data, then restoring three-class examples to that data should restore the behaviour, and it does: macro-F1 51.45 → 61.09, correct "maybe" answers 6 → 19 of 55, "no"→"yes" errors 63 → 26. The diagnosis in 5.3 was derived from the CPT-only adapters and predicted this result before it was run, which is the strongest evidence in the project that the mechanism is understood rather than curve-fitted.

The practical consequence is a change of plan for stage 3. The pipeline in Task 1.3 assumes alignment is a third training stage (DPO or similar). Here the reliability failure is created by a data-mixture choice in stage 2 and is recovered by correcting that choice, at a cost measured in Section 5.5. A preference-optimisation stage is therefore deprioritised in favour of C2, which traces the calibration-versus-retention trade-off. Section 5.6 shows that trade-off has a sharp knee: nearly all of the calibration gain arrives within the first 300 replay examples, while cost keeps rising with budget — so the practical recommendation is a small replay budget, with the caveat that the smallest budget tested over-corrects the very bias it was meant to fix — a pattern a second seed reproduces (Section 5.6).

The fix transfers across scale; the cause does not. The 35B check (Section 5.7, Figure 7) splits the argument above into a part that generalises and a part that does not. What generalises is the remedy: 300 balanced three-class examples in the SFT mix raise macro-F1 at both scales (+8.7 at 3B, +5.0 at 35B, two-seed means) and roughly triple the correct "maybe" answers, and at 35B they do so without the "yes"→"no" overshoot seen at 3B. What does not generalise is the diagnosis. At 3B the short-answer SFT data *creates* the bias ("no"→"yes" 42 → 63, "maybe" collapses); at 35B zero-replay SFT leaves the bias roughly where the base model already had it (the two seeds fall on either side of the base in macro-F1 and correct "maybe" answers, and "no"→"yes" rises by 2-4 against 21 at 3B), and the deficit is the zero-shot model's own reluctance to say "maybe" — 7 of 55 correct at both scales. Two practical consequences follow. First, the recommendation for stage 3 stands at both scales for this failure mode: correcting the SFT data mix is cheaper than a preference-optimisation stage and works whether the bias comes from the base model or from fine-tuning. Second, the cause has to be measured per model — the CPT-only-adapter diagnosis of Section 5.3 was only run at 3B, and at 35B the base model evaluation alone shows the problem predates SFT. The group-D ablations add that, at 35B, what goes into the data mix matters more than how the adapter is trained: only under-training moves macro-F1 outside the seed spread, and LoRA on attention layers alone (0.040 % of parameters) is enough. The replay budget curve transfers in shape but not in steepness (Figure 8): the smallest budget over-hedges at both scales and 300 is the best-balanced point at both, but at 35B every budget from 99 to 900 recovers most of the gain, where at 3B the gain was still growing between 99 and 300.

Continued pre-training does not help at either scale; what helps is in-domain instruction data. Five CPT variants have now been tested: PubMed abstracts (B1) and IU X-Ray reports (B2) at 3B, filtered and random CheXpert Plus reports (B3) at 3B, and 19.4 M tokens of textbooks, guidelines and encyclopedia text before instruction tuning at 35B (Section 5.14). The absence of a measurable improvement from knowledge pre-training holds across both SFT seeds at 35B: PathVQA closed falls slightly at both, while the seed-43 gains on PathVQA open and PneumoniaMNIST lack paired-interval support at seed 42; reasoning training has only a single-seed marginal gain. Every large improvement in the project instead came from putting a domain into the instruction data: SLAKE short answers (+10 closed, +19 open at 35B), PathVQA (+12.5), PneumoniaMNIST (+10.4), PubMedQA "maybe" replay (+5 macro-F1). The same holds for the free-text failure: descriptions improved after training on in-domain yes/no questions (S2-6datasets) or on reports from the same domain (S2-CheXpertPlus), not after knowledge pre-training. For a fixed GPU budget, the evidence here favours spending it on curated in-domain instruction data over a separate knowledge stage. Changing only the SFT seed shifts A's adult-film omission by about four points; the effect of Chinese dialogue remains unresolved because these results are keyword-judged and await CheXbert re-reading.

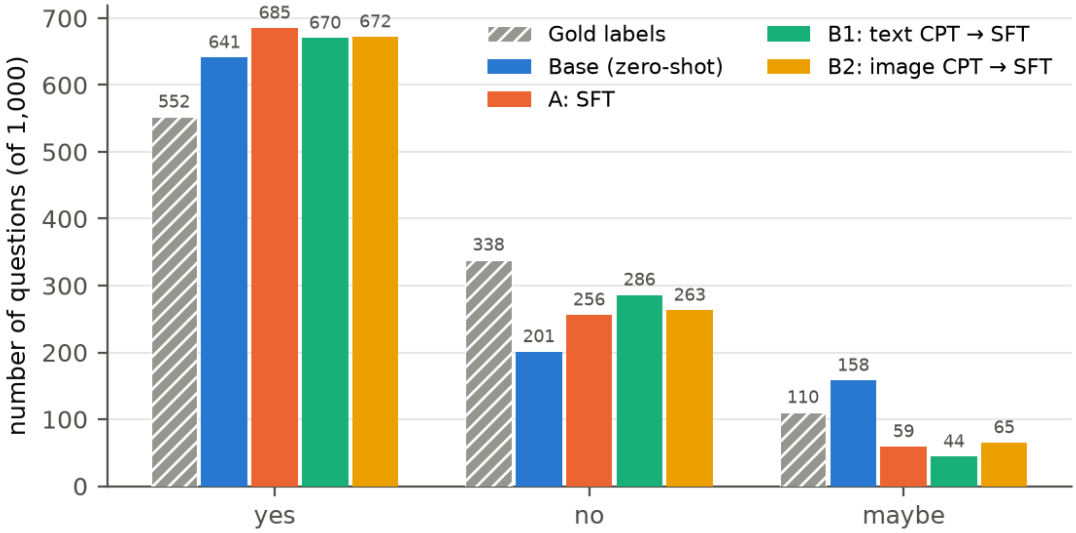


Figure 5. PubMedQA predicted-label counts versus gold. Each fine-tuning stage shrinks "maybe" further below its true frequency.

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7. Limitations and Planned Work

- The 3B experiments 0, A, B1, B2 and C1 are single runs, and no difference is tested statistically; differences under ±1 point are treated as noise. At 3B, the A-versus-B comparison (does a CPT stage help SFT?) was never repeated with a second seed: after C1 the remaining compute went to the replay budget and the 35B check instead. The B1/B2 conclusions are therefore stated as the absence of a detectable effect on SLAKE (every metric within ±0.6 of A, per-question changes balanced), not as evidence that CPT hurts; the drop on Abnormality-open questions (four of 41) is within what a single run can show. The A-server control partly substitutes for a second seed on experiment A: an independent run with different hardware, precision and quantisation reproduced every metric to within half a point. The replay-budget comparison has two seeds at both 300 and 900 (Section 5.6): run-to-run spread is under 0.7 macro-F1, up to about 5 points of single-class recall and up to 7 "maybe" questions; the 300-versus-900 gap in "yes" recall exceeds the larger spread, and the 99-example point extends the same trend. The zero point and the 99 point remain single runs.
- The PubMedQA "maybe" class has only 55 unique training items, repeated about five times to reach 300 in C1's balanced sample. The calibration gain may therefore depend partly on memorising a small set; the held-out half shows the effect transfers, but a larger three-class source would test it properly. On the training half C1 scores 94.4 % accuracy and answers 54 of 55 "maybe" questions correctly, which confirms the memorisation is present and is exactly why the split protocol exists. The 99-example point bounds the concern: with each "maybe" item seen at most once, "maybe" recall is the highest of any run (41.8 %), so restoring the class does not depend on repetition; what the larger budgets add is discrimination (accuracy 64.6 → 69.6 → 73.0), not the class itself.
- C1's general-ability cost on TextVQA is −2.23 points on 300 questions (about seven). A second probe (MMBench, 500 multiple-choice) shows no replay cost, so the TextVQA figure should be read as a short-answer-format effect, not as general forgetting. At 3B neither probe covers open-ended description; the free-text probe was added only for the 35B models (Section 5.7), where it shows no drift in long-form general description.
- The 35B check (Section 5.7) differs from the 3B runs in three forced ways: custom training code instead of LLaMA-Factory, LoRA on attention and the shared expert only (the routed experts are packed parameters), and evaluation with thinking disabled. Run-to-run spread is larger at 35B (up to 3.7 macro-F1 between seeds); the four replay budgets (0 / 99 / 300 / 900) and the attention-only and one-epoch settings have two seeds, the other ablations one. Thinking mode was measured once, on the base model only, at a 4,096-token budget that 21 % of answers exceeded; whether thinking combines with replay is not measured. The free-text probe's medical false-normal count proved too noisy over 44 images to support any conclusion. The external test sets (Section 5.8) were run on five 35B models only (no 3B, two seeds per condition); free-text abnormality reporting on PneumoniaMNIST is judged by a keyword rule validated by hand on 50 descriptions, and remains seed-dependent even over 390 abnormal images, so a model-based judge with human review and more seeds is needed before it can rank models. PneumoniaMNIST is paediatric and 224×224.
- Evaluation uses greedy decoding and string matching; open-ended recall rewards verbose answers. Exact match and F1 are reported alongside to bound this.

- The vision tower is frozen throughout. Unfreezing it (or LoRA on the ViT) is a natural ablation for B2 but roughly doubles memory.
- The missed-abnormality measure of Section 5.10 rests on automatic CheXbert labels and a keyword judge (validated by hand on 30 descriptions; the per-finding naming rates are lenient and unvalidated), has only 231 normal films, and covers five 35B models with one decoding each; the free-text rates vary strongly between training seeds.
- Image-text CPT was tested on IU X-Ray (B2, 3.5k pairs) and CheXpert Plus (B3, 5k pairs per arm), always with a frozen vision tower and one epoch; neither helped the main task. B3 has one seed per arm, arms that share half their data, and a weak alignment scorer (1.9 % R@1), so it cannot rule out a small effect of data-quality filtering; a direct report-generation comparison of the two CPT adapters is the planned follow-up.
- S2-6datasets, S2-CheXpertPlus and the three-stage model C (Sections 5.12–5.14) remain single runs. A and B now have two SFT seeds with paired bootstrap comparisons, both supporting no reproducible gain from knowledge pre-training, but B43 reuses the seed-42 stage-1 adapter: variability of pre-training itself is not tested. A's adult-film omission changes by about four points between seeds, so the effect of Chinese dialogue remains unresolved. CheXbert re-reading of A42, B42, A43, B43, C and the stage-1 adapter is pending; their adult-film omission rates are keyword-judged only, with no paired intervals computed because per-image labels stay on the server.
- The three-stage comparison tests one corpus size (19.4 M tokens, one epoch) and LoRA on attention and the shared expert only. A null result for knowledge pre-training at this scale does not exclude an effect from a much larger corpus or from full-parameter pre-training. MedQA and CMExam are near their ceiling for this model (85–86 % and 93–94 %), so they cannot show a large gain from reasoning training; a harder Chinese exam set would be needed.
- Kaggle's 30 h/week GPU quota bounds throughput at roughly three full experiments per week; the college Docker GPU server, once available, removes the 4-bit requirement (bf16, larger batch).

---

8. Reproducibility

Model card: `docs/cards/MODEL_CARD.md` (adapters, hyper-parameters, results, intended use, measured failure modes). Data card: `docs/cards/DATA_CARD.md` (every dataset, the PubMedQA split protocol, the IU X-Ray composition bias, the scoring correction, compliance rules).

```
git clone https://github.com/AugustLoo/MedLoRA && cd MedLoRA
pip install -r requirements.txt
pip install "llamafactory[torch,metrics] @ git+https://github.com/hiyouga/LLaMA-Factory.git"

python data/download_slake.py && python data/convert_slake_sharegpt.py
python data/download_pubmedqa.py && python data/convert_pubmedqa_cpt.py --max 10000
python data/convert_iu_xray.py --root <IU_XRAY_DIR> --max 5000      # B2 only

bash train/eval_all.sh baseline                                       # experiment 0
llamafactory-cli train configs/sft_slake_qlora.yaml                   # A
bash train/eval_all.sh sft_r16 outputs/sft_slake_qlora_r16
llamafactory-cli train configs/cpt_pubmed_qlora.yaml                  # B1
llamafactory-cli train configs/sft_after_cpt.yaml
bash train/eval_all.sh cpt_sft_r16 outputs/sft_after_cpt_r16
```

```
llamafactory-cli train configs/cpt_iu_qlora.yaml                    # B2
llamafactory-cli train configs/sft_after_cpt_iu.yaml
bash train/eval_all.sh cpt_iu_sft_r16 outputs/sft_after_cpt_iu_r16
```

Every run is fully specified by one YAML (seed 42) and one evaluation tag; the JSON summaries in `results/` are the exact files behind Section 5. Kaggle notebooks that execute the sequences above end-to-end are under `notebooks/`.

Experiments C1 and C2 (replay; bf16 on a single 24-32 GB GPU, configs under `configs/bf16/`). The PubMedQA split is fixed in `data/pubmedqa_split.json`; only its train half is ever converted into training data.

```
python data/convert_pubmedqa_sft.py --per-class 300                 # C1, 900 examples
python data/convert_pubmedqa_sft.py --per-class 100 --tag 300       # C2, 300 examples
python data/convert_pubmedqa_sft.py --per-class 100 --tag 300s43 --seed 43   # C2-300, second seed
llamafactory-cli train configs/bf16/sft_slake_qlora.yaml            # A-server (replay 0)
llamafactory-cli train configs/bf16/sft_mix_pubmedqa_300.yaml       # likewise _100, _300_s43, _900_s43, sft_m
ix_pubmedqa (C1)
bash train/eval_all.sh sft_mix_300 outputs/sft_mix_pubmedqa_300_r16
python scripts/eval_pubmedqa_split.py                               # every model re-scored on the held-out ha
lf
```

35B scaling check (Section 5.7; four 32 GB GPUs; full runbook, environment pins and host rules in `docs/INTERNS2.md`). The training environment needs `transformers==5.2.0`; serving uses LMDeploy (TP 4) with thinking disabled, and `serve.sh` expects an LMDeploy environment at the `ENV_DIR` path set at the top of the script.

```
CUDA_VISIBLE_DEVICES=0,1,2,3 python train/interns2/train_lora.py \
    --data data/processed/slake_train.json data/processed/pubmedqa_sft_train_300.json \
    --batch 4 --accum 4 --output outputs/interns2_mix_300            # replay 0: SLAKE file only
    # add --path-map OLD=NEW if the image paths inside the JSON point to another machine
CUDA_VISIBLE_DEVICES=0,1,2,3 python train/interns2/merge_lora.py \
    --adapter outputs/interns2_mix_300 --out <MERGED_DIR>
bash train/interns2/serve.sh start <MERGED_DIR>                      # OpenAI-compatible server on port 23334
MEDVLM_API_BASE=http://127.0.0.1:23334/v1 \
MEDVLM_API_EXTRA='{"chat_template_kwargs":{"enable_thinking":false}}' \
MODEL=<MERGED_DIR> bash train/eval_all.sh interns2_mix_300
python eval/eval_openended.py --model <MERGED_DIR> --tag interns2_mix_300   # free-text probe (same env vars)
bash train/interns2/serve.sh stop
OPENENDED=1 bash train/interns2/ablation.sh s43 r0_s43 attn ep1          # second seeds and group D, fully autom
ated
bash train/interns2/ablation.sh r900 r100 r900_s43 r100_s43              # replay budget curve (data: docs/INTE
RNS2.md)
python scripts/make_fig7_scale.py && python scripts/make_fig8_dose.py    # Figures 7 and 8
bash train/interns2/external.sh prefetch                                 # external test sets (Section 5.8), tes
t shards only
bash train/interns2/external.sh base final r0 r0_s43 s43                 # VQA-RAD, PathVQA, MedQA, PneumoniaMNI
ST
python scripts/rescore_pneumonia.py                                      # free-text abnormality judge (validate
```

```
d 2026-10-01)
# B3 (user0 container; CheXpert Plus stays on the server; the teammate's code is used read-only)
python scripts/score_chexpert_pairs.py --split val --retrieval          # reproduce the scorer's validation R@1
(0.0191)
python scripts/build_b3_pool.py verify && python scripts/build_b3_pool.py sample && python scripts/build_b3_pool.
py convert
python scripts/score_chexpert_pairs.py --pairs <pool manifest> --image-cache-root <pool JPGs> --split b3pool
python data/convert_chexpert_b3.py && bash train/run_b3.sh             # top-5k vs random-5k CPT, SLAKE SFT, fou
r tables
# Missed abnormalities (Section 5.10): host serves one 35B model at a time, the container evaluates
bash train/interns2/serve_for_container.sh base                          # host; also final / r0 / r0_s43 / s43
python eval/eval_chexpert.py --model <served model dir> --tag interns2_base --concurrency 8   # container
python eval/chexbert_judge.py --help                                     # container, CPU: CheXbert re-reading (S
ection 5.11)
# S2-6datasets (Section 5.12) and S2-CheXpertPlus (Section 5.13), same settings as replay 300, data only changes
bash train/interns2/run_6ds.sh data && bash train/interns2/run_6ds.sh train && bash train/interns2/run_6ds.sh mer
ge
bash train/interns2/run_6ds.sh eval && bash train/interns2/run_6ds.sh external
python data/convert_chexpert_s2.py                                        # container; then copy to the host
bash train/interns2/run_cxp.sh data && bash train/interns2/run_cxp.sh train   # likewise merge / eval / external
# Three-stage training (Section 5.14; plan and data sources in docs/PLAN_35B_CPT.md)
bash train/interns2/run_cpt35.sh data && bash train/interns2/run_cpt35.sh sweep && bash train/interns2/run_cpt35.
sh train
bash train/interns2/run_sft35.sh data2 && bash train/interns2/run_sft35.sh data3
bash train/interns2/run_sft35.sh train a     # also b (from stage 1) and c (from b); a43 / b43 = seed 43
bash train/interns2/run_sft35.sh merge a && bash train/interns2/run_sft35.sh eval a    # English, Chinese, no-ima
ge, external
python scripts/bootstrap_ci.py --help && python scripts/summarize_3stage.py           # paired intervals, snapsh
ot
python scripts/summarize_seed43.py                                      # A/B across SFT seeds 42 and 43
```

References

BibTeX entries: `docs/refs.bib` (keys in back-quotes below, cited in the text as `\citep{key}`).

> **Verification.** The 29 original entries were checked on 2026-09-23 against the arXiv abstract page or the publisher page. Titles, first authors, identifiers and years all matched; three entries were corrected (MMBench venue, Model Cards entry type, Med-Flamingo author order) and seven were completed with volume, page or DOI fields. The nine entries added for Section 5.7 were checked on 2026-09-27, the two added for Section 5.8 on 2026-10-01, and the nine added for Sections 5.11–5.14 on 2026-10-24, against arXiv, the ACL Anthology, the Hugging Face model pages and the LMDeploy repository. Details in the header of `docs/refs.bib`.

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